Experiment-6

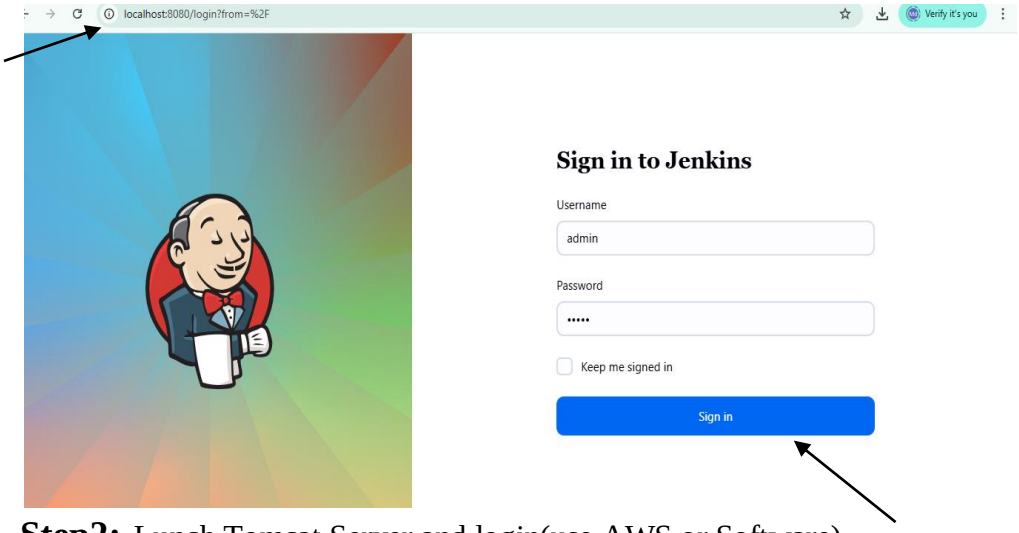
AIM:-Deploy the artifact on the Test Server

Softwares:Jenkins(port:8080),JDk 17 or 21,Tomcat 9 (port:9090),Gitbash

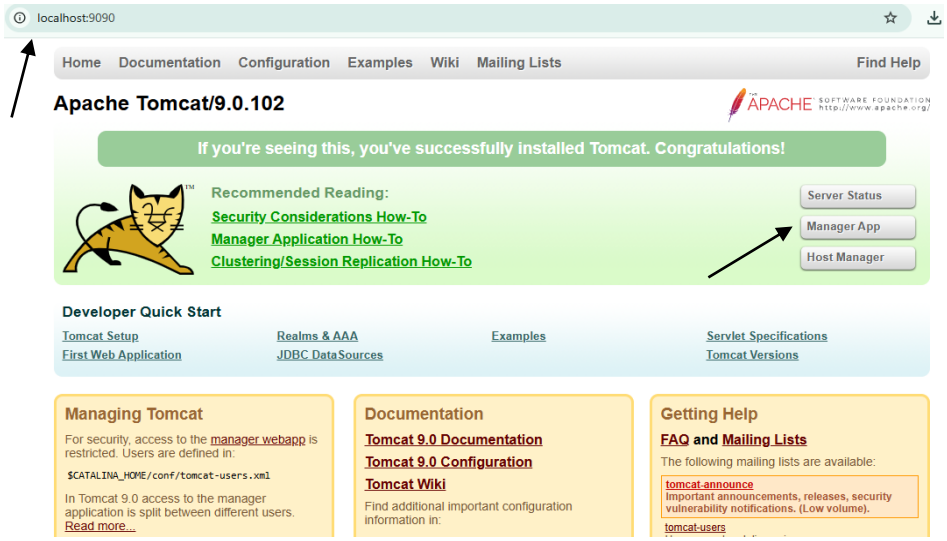
(Here we are creating one artifact war file. That war file we are deploy to test server(Tomcat server) using Jenkins Server)

(**Note:-**Images are purpose of understanding only don't paste it in records but output images paste it in output area)

Step1:-Lunch Jenkins and Login(Use AWS or Software)



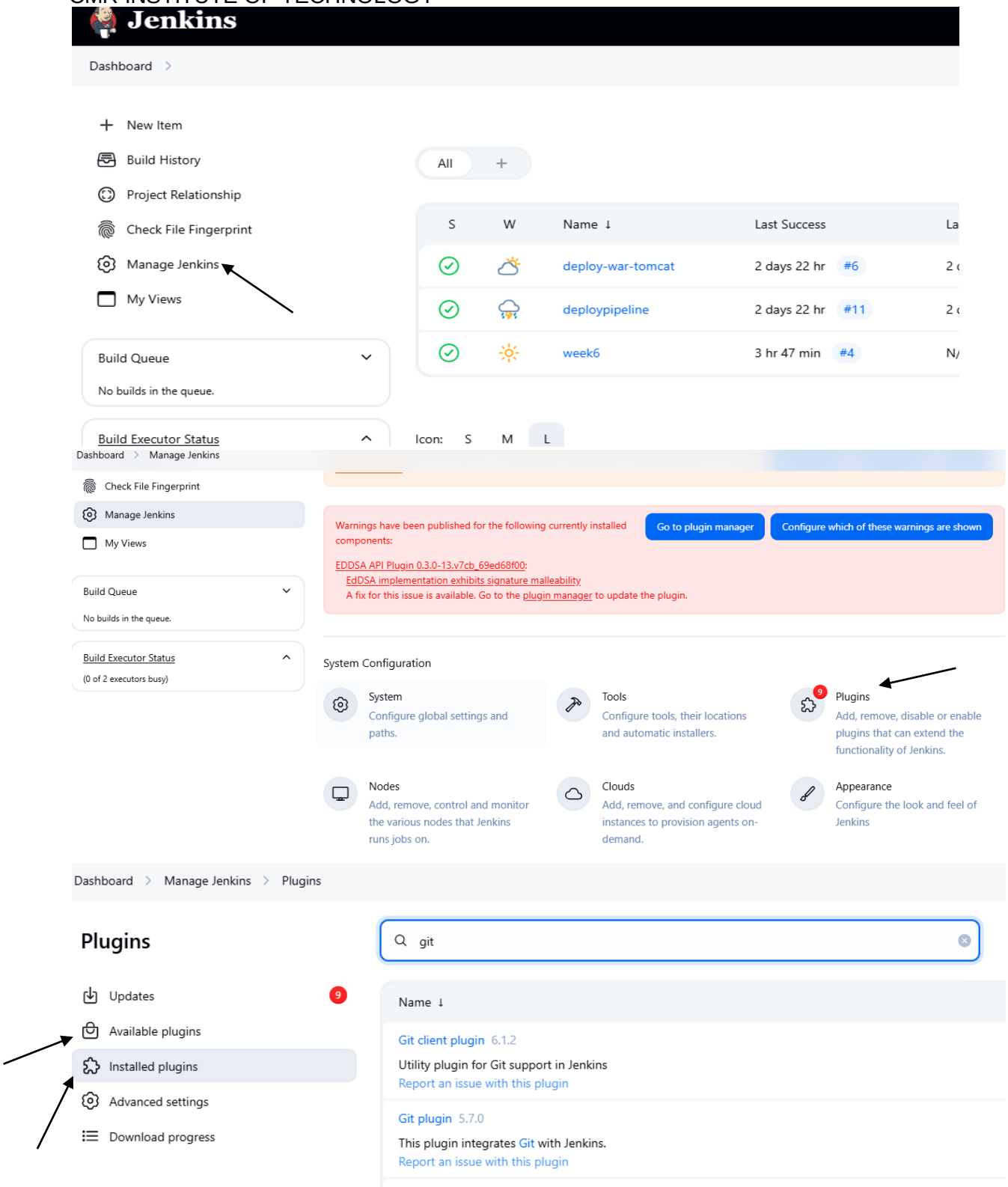
Step2:-Lunch Tomcat Server and login(use AWS or Software)



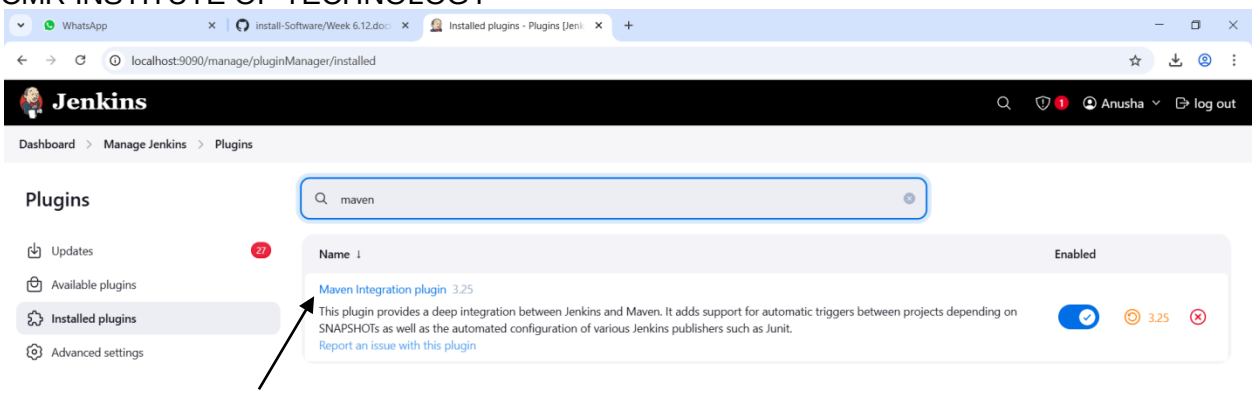
Open Jenkins do following steps

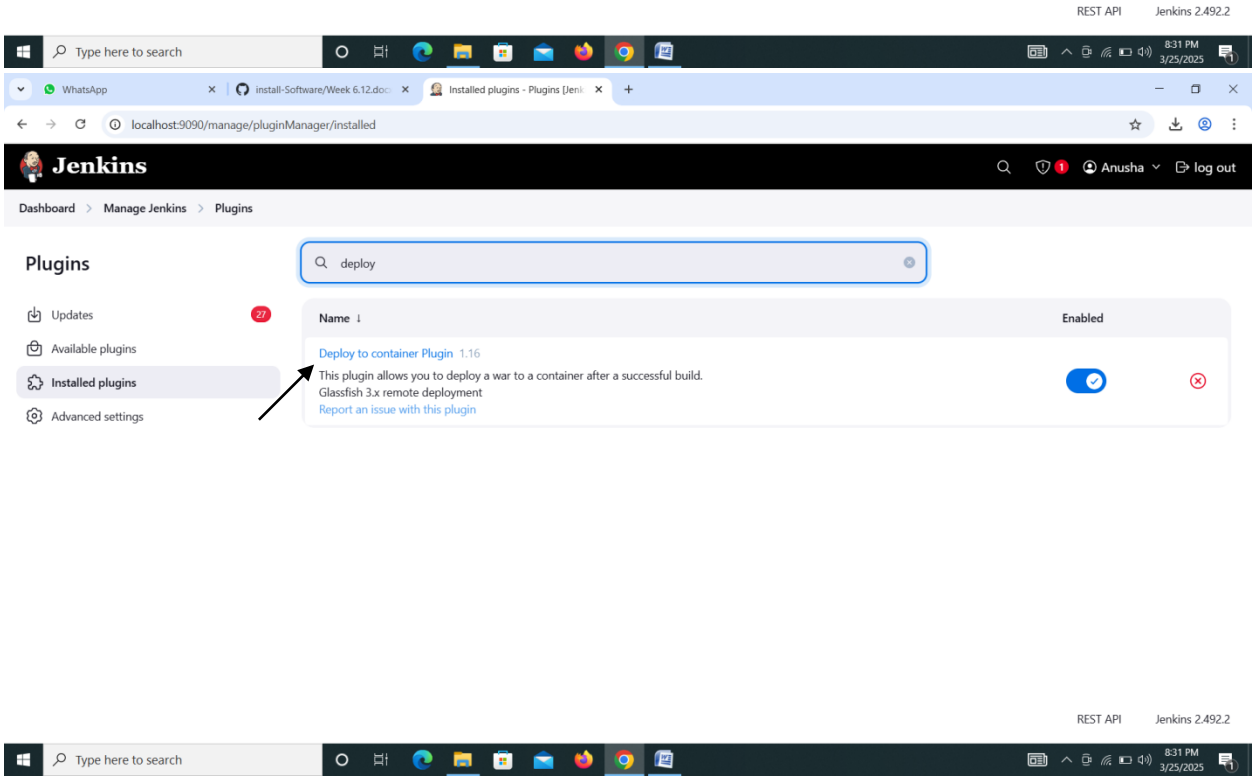
Step 3:-Install plugins

Go to Manage Jenkins→Plugins→Install Plugin select verify Git→Go to Available plugins install Maven integration and Deploy to container install.



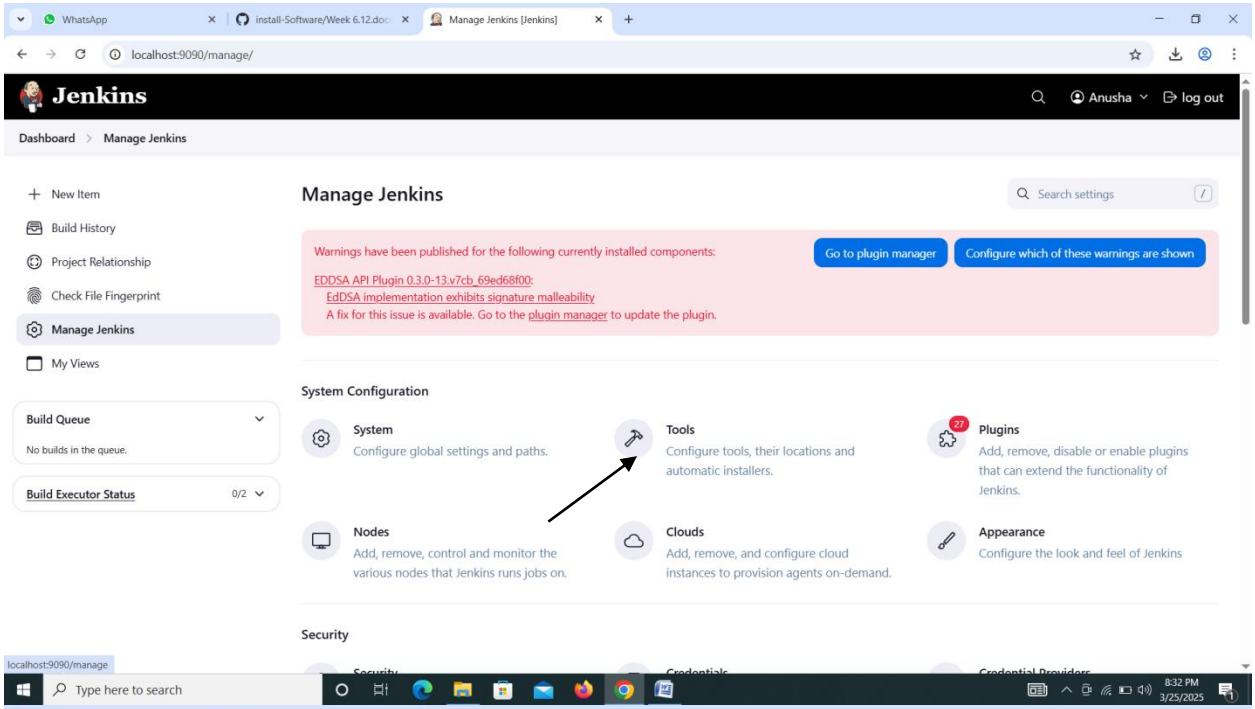
Above picture three plugins are installed , If not showing please go to available plugins install it.

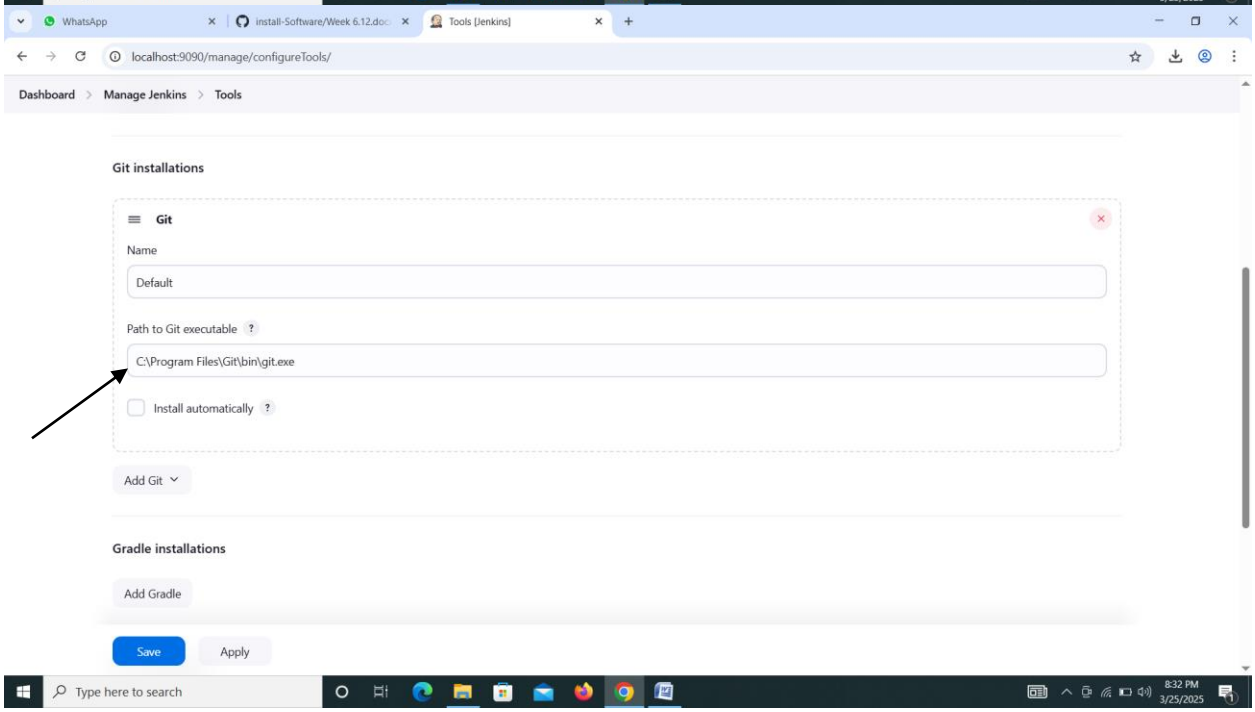


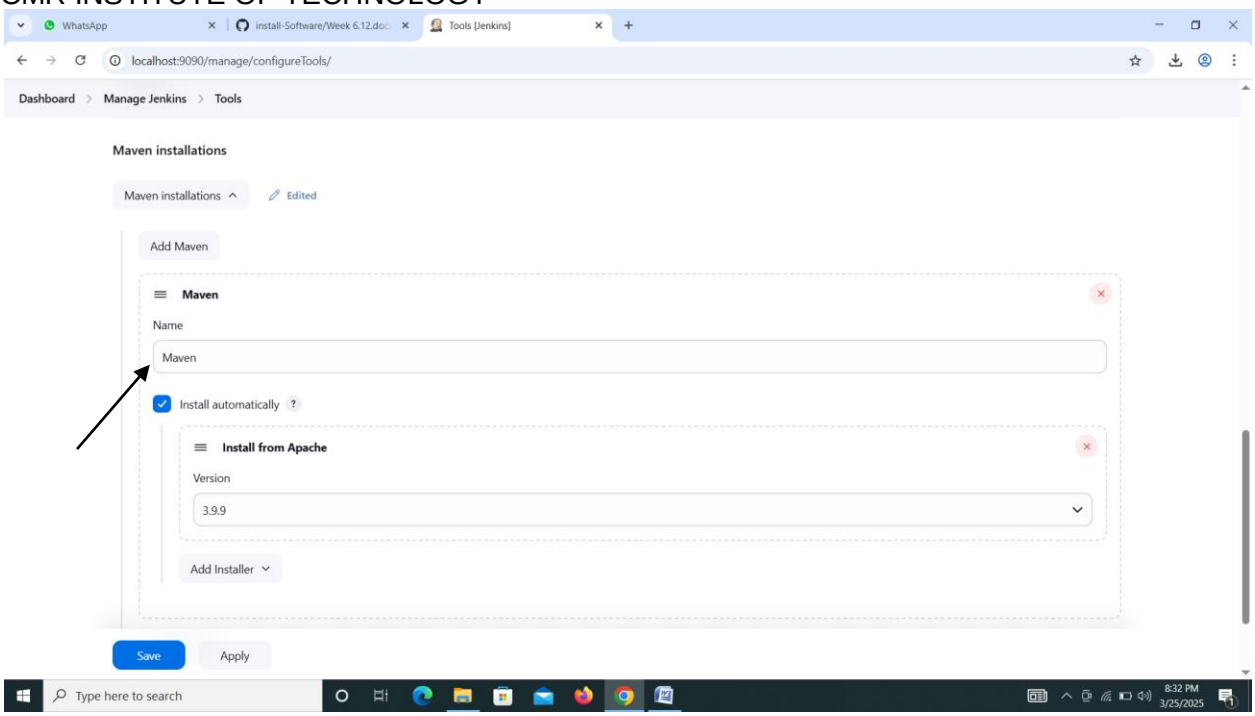


Step 4:-set up git path and maven installation in Tools

Go to manage Jenkins→Tools→In git installation select path(Git bash) if your using Jenkins software or AWS leave it →In maven Enter name Maven and default installation→Apply and Save.

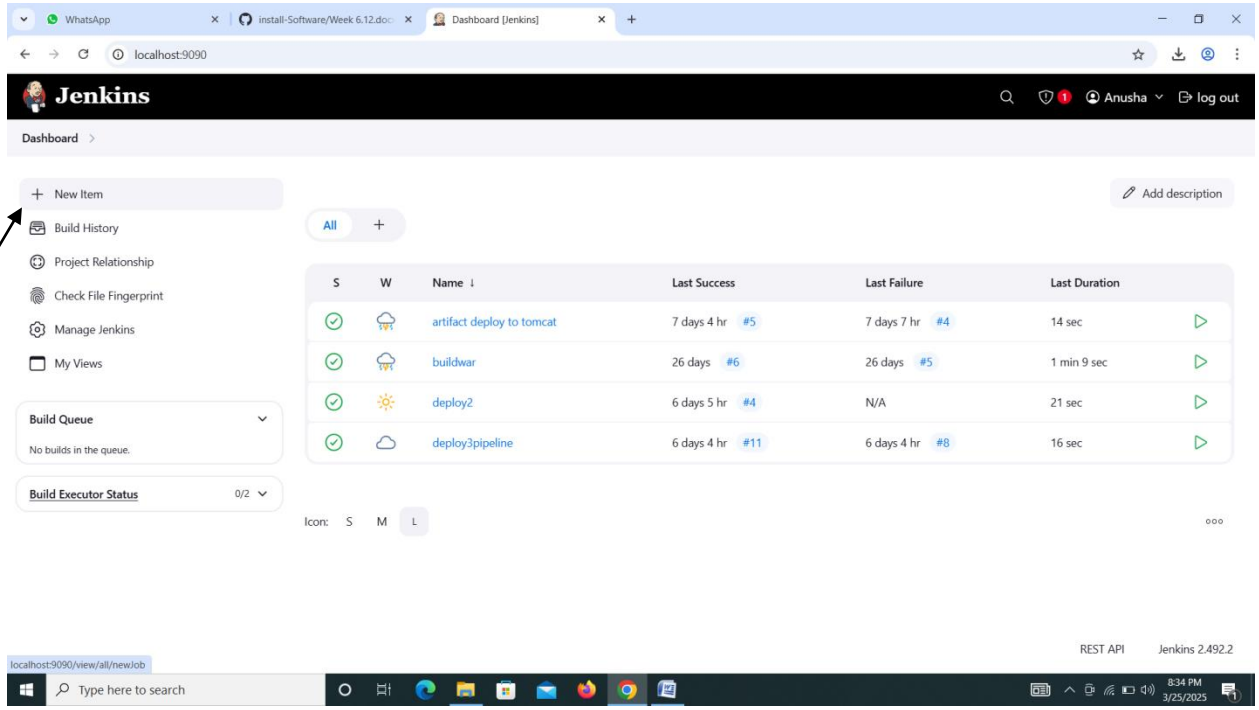

Manage Jenkins [Jenkins]
localhost:9090/manage/
Jenkins
Anusha
log out
Dashboard > Manage Jenkins
New Item
Build History
Project Relationship
Check File Fingerprint
Manage Jenkins
My Views
Build Queue
No builds in the queue.
Build Executor Status 0/2
Manage Jenkins
Search settings
Warnings have been published for the following currently installed components:
Go to plugin manager
Configure which of these warnings are shown
EDDSA API Plugin 0.3.0-13.v7cb_69ed68f00:
EdDSA implementation exhibits signature malleability
A fix for this issue is available. Go to the plugin manager to update the plugin.
System Configuration
System
Configure global settings and paths.
Tools
Configure tools, their locations and automatic installers.
Plugins
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.
Nodes
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
Clouds
Add, remove, and configure cloud instances to provision agents on-demand.
Appearance
Configure the look and feel of Jenkins
Security

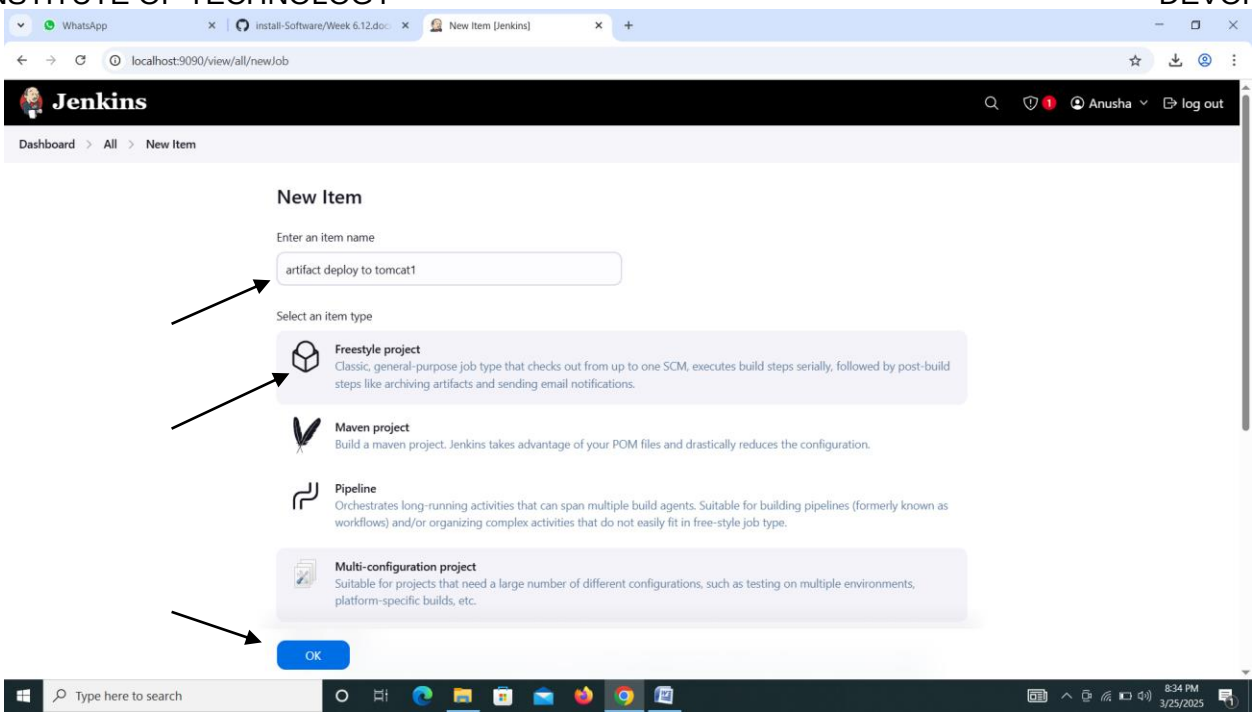

Tools [Jenkins]
localhost:9090/manage/configureTools/
Dashboard > Manage Jenkins > Tools
Git installations
Git
Name
Default
Path to Git executable
C:\Program Files\Git\bin\git.exe
Install automatically
Add Git
Gradle installations
Add Gradle
Save
Apply



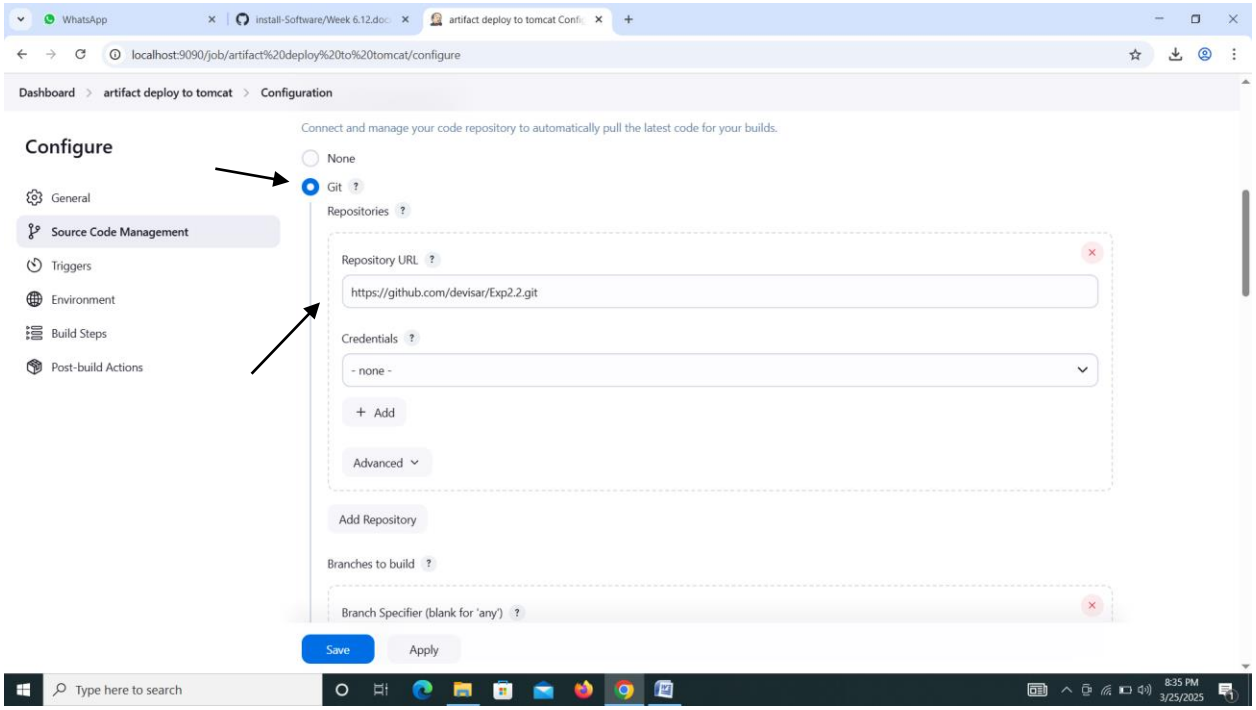
Step 5:-Creating New Job for deploy artifact in tomcat test server

1. Create New Item→ Enter name DeploytoTestServer →select freestyle project→click ok

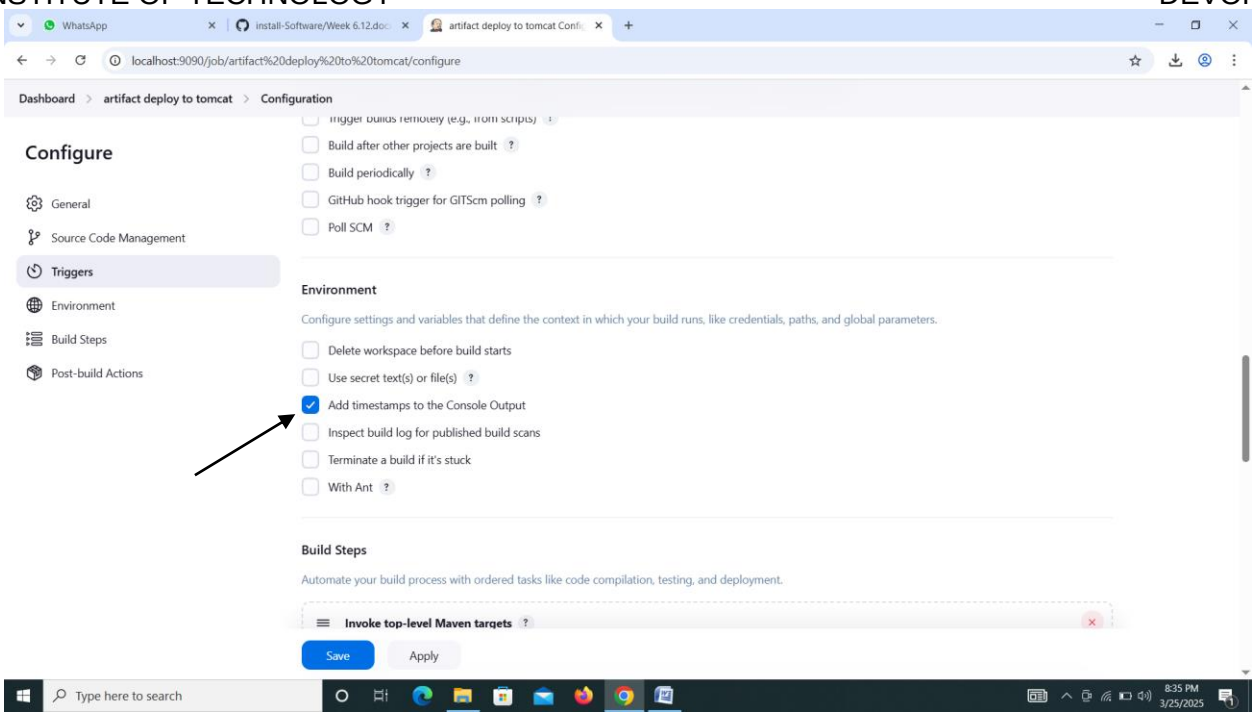




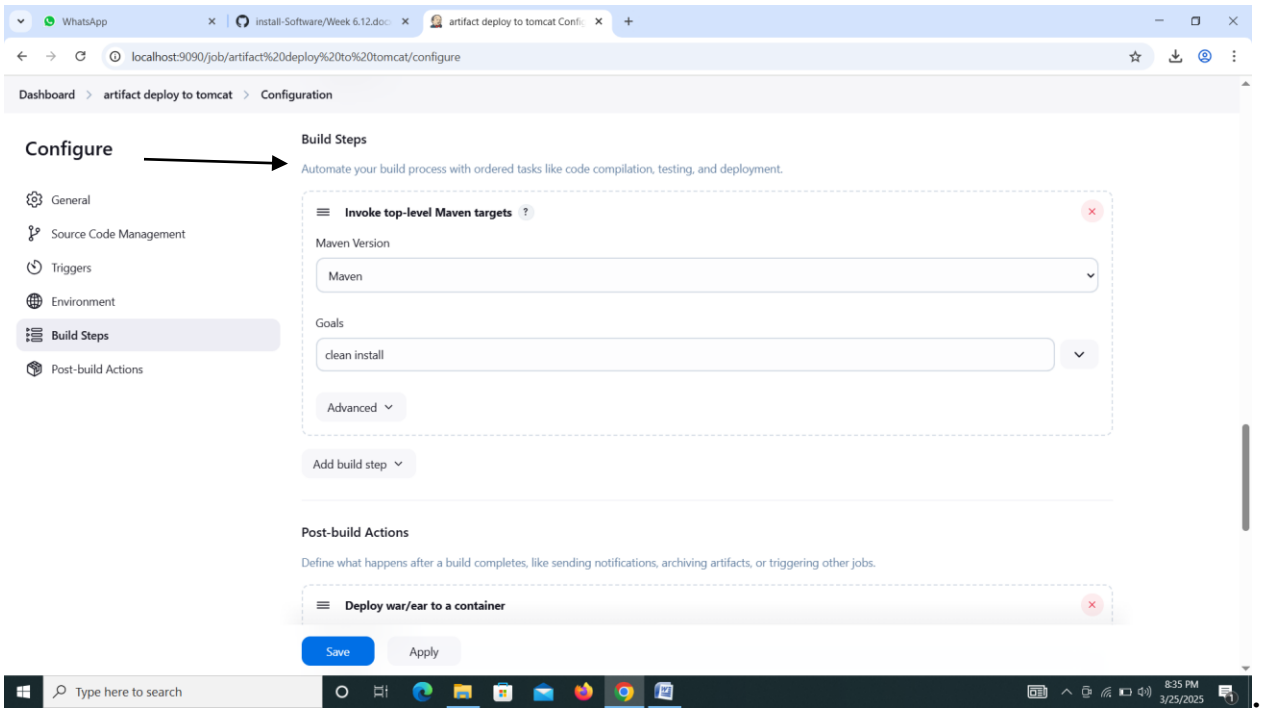
2. Select git →paste maven project url form Github→Must and should match branches



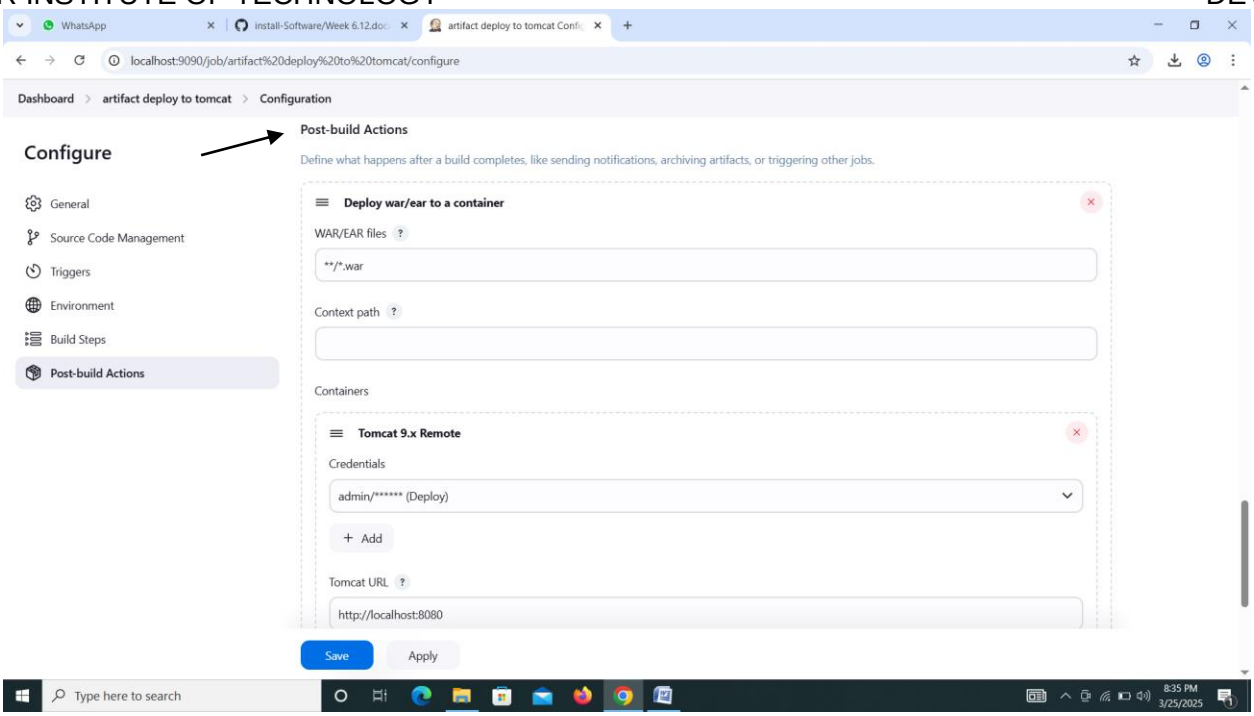
3. Go to Environment →Select Add timestamps to the Console Output

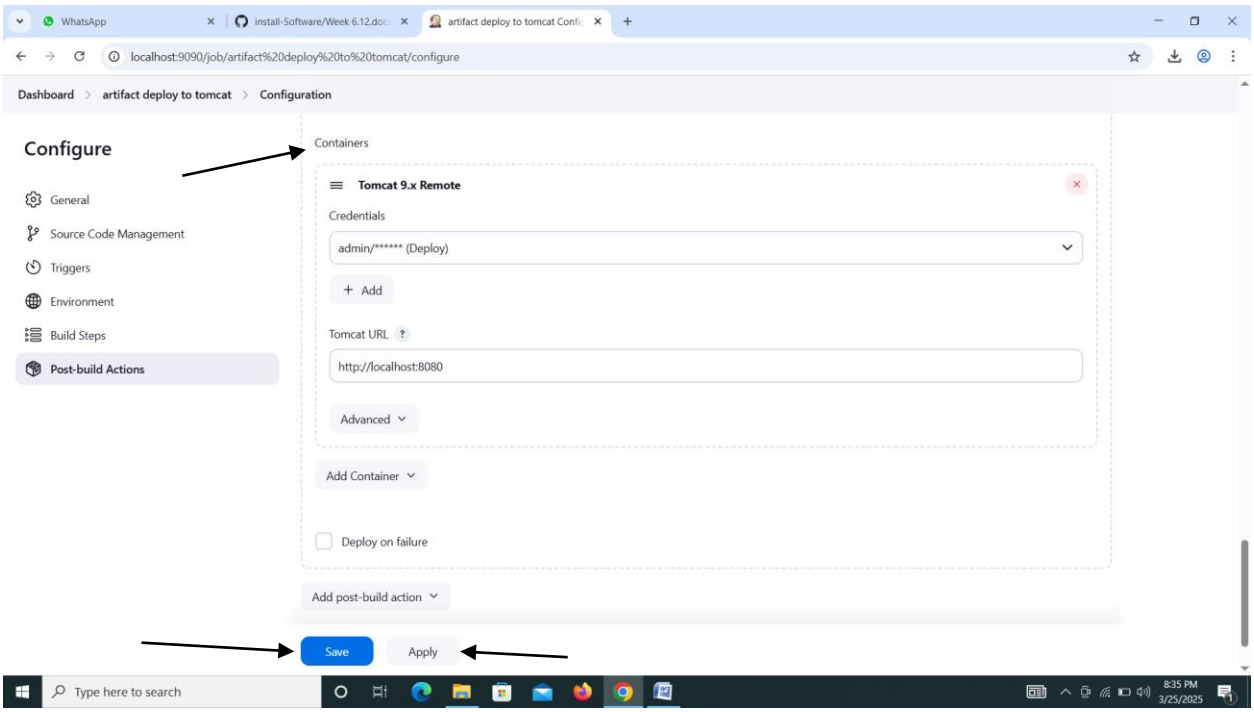


4. Go to Build Setups→Select **Invoke top-level Maven targets→Maven Version enter maven →Goals clean install**



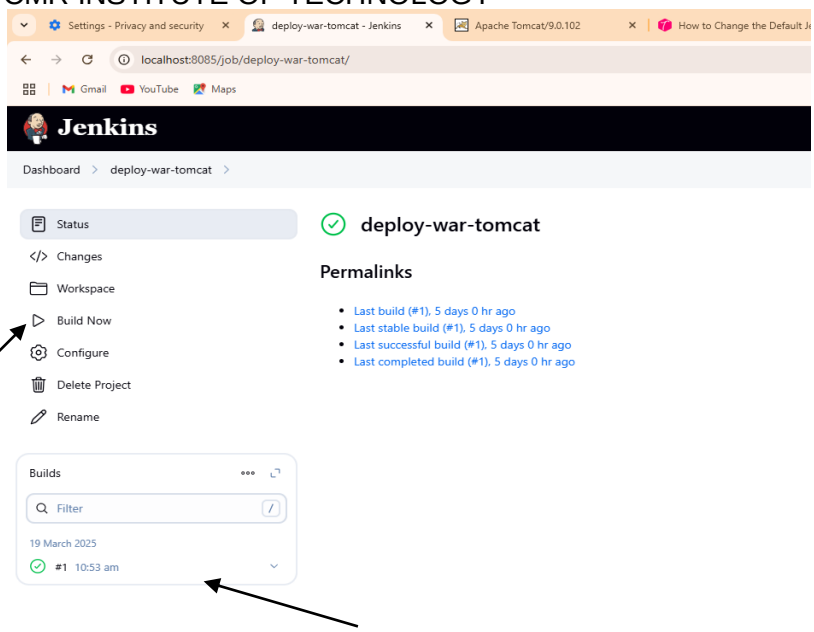
5. **Go to** Post-build Actions→Select **Deploy war/ear to a container→**WAR/EAR files enter **/*.war→select Containers **Tomcat 9.x Remote→Add credentials like username, password, Id and description.-->Tomcat URL http://localhost:9090 →Apply and save.**

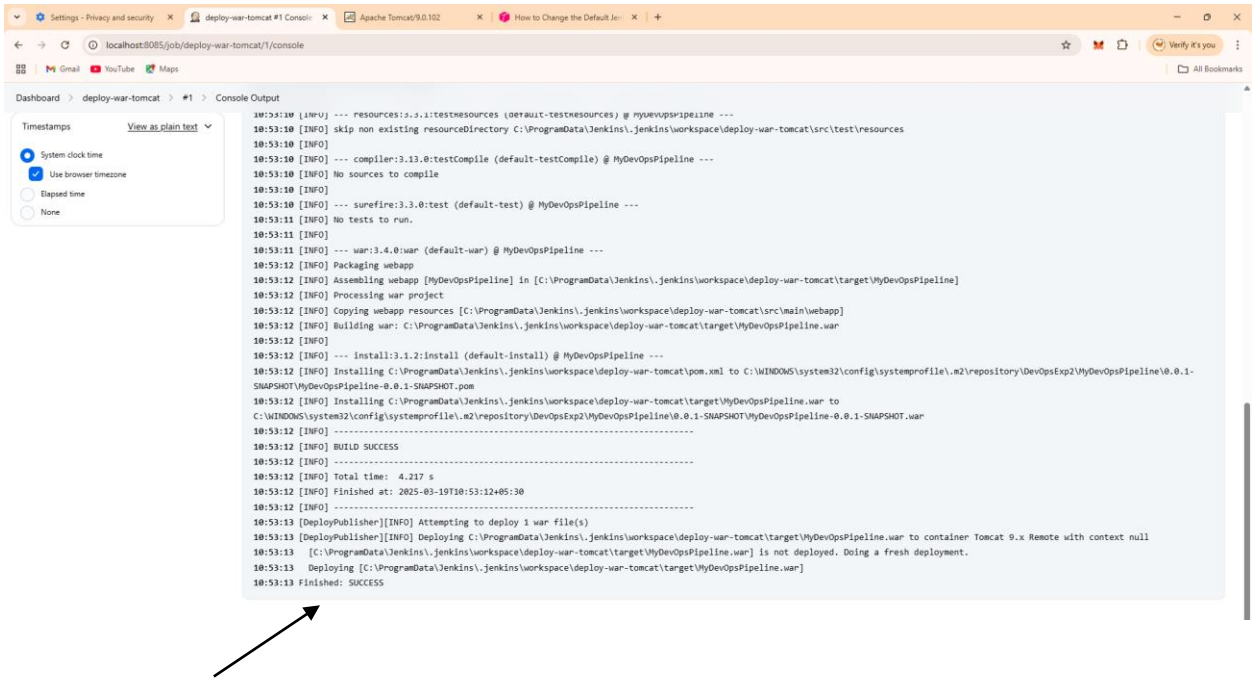




Step 6:- Build war File

1. Click Build now →Go to Build select console output now you got Successfully Finish.


Settings - Privacy and security
deploy-war-tomcat - Jenkins
Apache Tomcat/9.0.102
How to Change the Default J
localhost:8085/job/deploy-war-tomcat/
Gmail
YouTube
Maps
Jenkins
Dashboard > deploy-war-tomcat >
Status
Changes
Workspace
Build Now
Configure
Delete Project
Rename
Builds
Filter
19 March 2025
#1 10:53 am
deploy-war-tomcat
Permalinks
Last build (#1), 5 days 0 hr ago
Last stable build (#1), 5 days 0 hr ago
Last successful build (#1), 5 days 0 hr ago
Last completed build (#1), 5 days 0 hr ago


Settings - Privacy and security
deploy-war-tomcat #1 Consol
Apache Tomcat/9.0.102
How to Change the Default Jen
localhost:8085/job/deploy-war-tomcat/1/console
Verify it's you
Gmail
YouTube
Maps
All Bookmarks
Dashboard > deploy-war-tomcat > #1 > Console Output
Timestamps
View as plain text
System clock time
Use browser timezone
Elapsed time
None
10:53:10 [INFO] --- resources:3.3.1:testResources (default-testResources) @ MyDevOpsPipeline ---
10:53:10 [INFO] skip non existing resourceDirectory C:\ProgramData\Jenkins\.jenkins\workspace\deploy-war-tomcat\src\test\resources
10:53:10 [INFO]
10:53:10 [INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ MyDevOpsPipeline ---
10:53:10 [INFO] No sources to compile
10:53:10 [INFO]
10:53:10 [INFO] --- surefire:3.3.0:test (default-test) @ MyDevOpsPipeline ---
10:53:11 [INFO] No tests to run.
10:53:11 [INFO]
10:53:11 [INFO] --- war:3.4.0:war (default-war) @ MyDevOpsPipeline ---
10:53:12 [INFO] Packaging webapp
10:53:12 [INFO] Assembling webapp [MyDevOpsPipeline] in [C:\ProgramData\Jenkins\.jenkins\workspace\deploy-war-tomcat\target\MyDevOpsPipeline]
10:53:12 [INFO] Processing war project
10:53:12 [INFO] Copying webapp resources [C:\ProgramData\Jenkins\.jenkins\workspace\deploy-war-tomcat\src\main\webapp]
10:53:12 [INFO] Building war: C:\ProgramData\Jenkins\.jenkins\workspace\deploy-war-tomcat\target\MyDevOpsPipeline.war
10:53:12 [INFO]
10:53:12 [INFO] --- install:3.1.2:install (default-install) @ MyDevOpsPipeline ---
10:53:12 [INFO] Installing C:\ProgramData\Jenkins\.jenkins\workspace\deploy-war-tomcat\pom.xml to C:\WINDOWS\system32\config\systemprofile\.m2\repository\DevOpsExp2\MyDevOpsPipeline\0.0.1-SNAPSHOT\MyDevOpsPipeline-0.0.1-SNAPSHOT.pom
10:53:12 [INFO] Installing C:\ProgramData\Jenkins\.jenkins\workspace\deploy-war-tomcat\target\MyDevOpsPipeline.war to C:\WINDOWS\system32\config\systemprofile\.m2\repository\DevOpsExp2\MyDevOpsPipeline\0.0.1-SNAPSHOT\MyDevOpsPipeline-0.0.1-SNAPSHOT.war
10:53:12 [INFO] --
10:53:12 [INFO] BUILD SUCCESS
10:53:12 [INFO] --
10:53:12 [INFO] Total time: 4.217 s
10:53:12 [INFO] Finished at: 2025-03-19T10:53:12+05:30
10:53:12 [INFO] --
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10:53:13 [DeployPublisher][INFO] Deploying C:\ProgramData\Jenkins\.jenkins\workspace\deploy-war-tomcat\target\MyDevOpsPipeline.war to container Tomcat 9.x Remote with context null
10:53:13 [C:\ProgramData\Jenkins\.jenkins\workspace\deploy-war-tomcat\target\MyDevOpsPipeline.war] is not deployed. Doing a fresh deployment.
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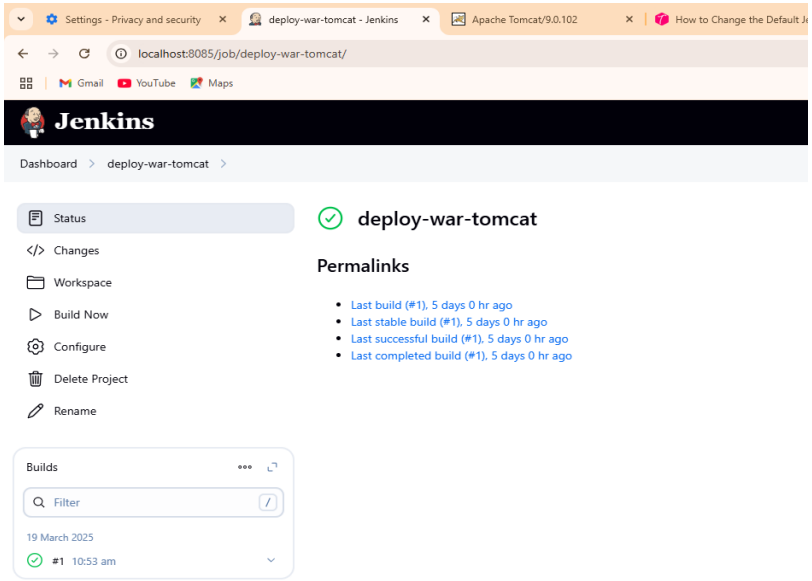
Output:-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YouTube
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Jenkins
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Status
Changes
Workspace
Build Now
Configure
Delete Project
Rename
Builds
Filter
19 March 2025
#1 10:53 am
deploy-war-tomcat
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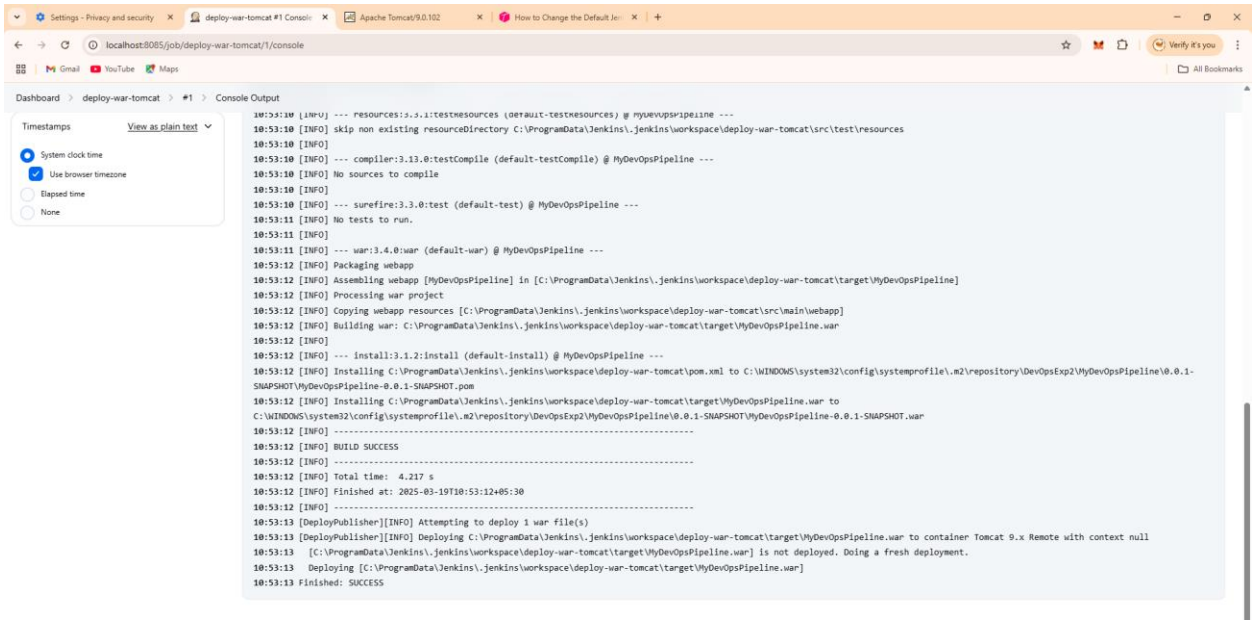
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Maps
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View as plain text
System clock time
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10:53:12 [INFO] Installing C:\ProgramData\Jenkins\.jenkins\workspace\deploy-war-tomcat\target\MyDevOpsPipeline.war to C:\WINDOWS\system32\config\systemprofile\.m2\repository\DevOpsExp2\MyDevOpsPipeline\0.0.1-SNAPSHOT\MyDevOpsPipeline-0.0.1-SNAPSHOT.war
10:53:12 [INFO] --
10:53:12 [INFO] BUILD SUCCESS
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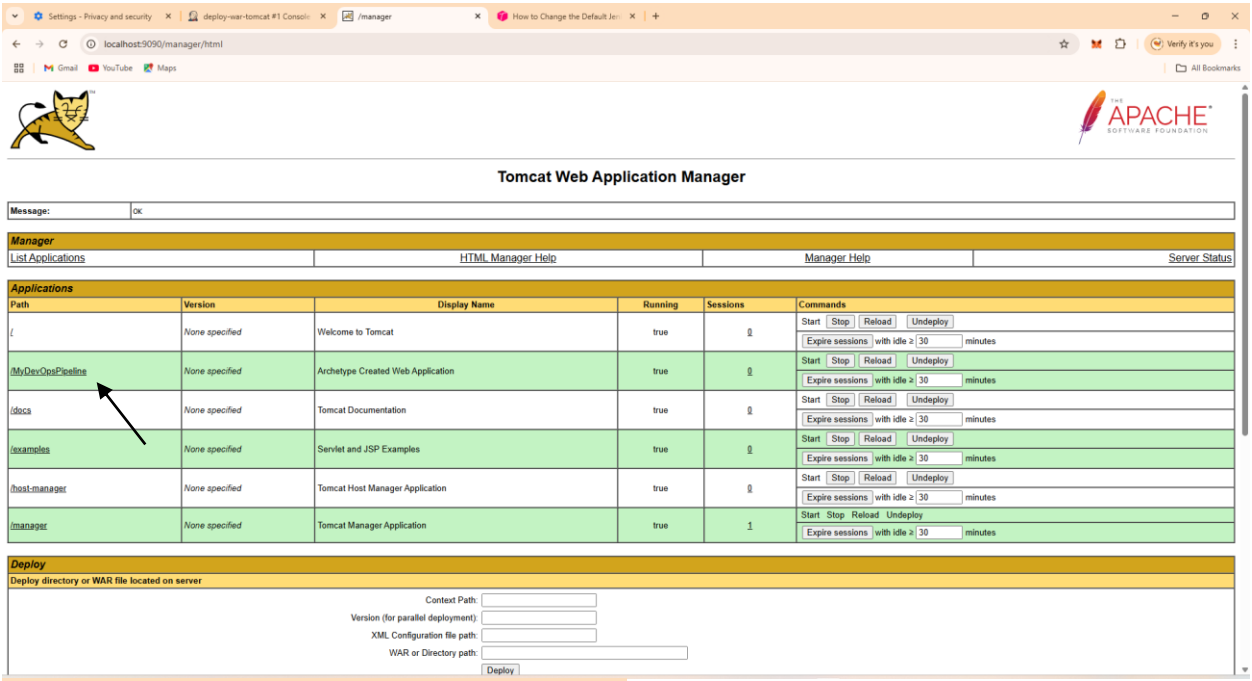
10:53:13 Finished: SUCCESS



Hello Students12345

Experiment-7

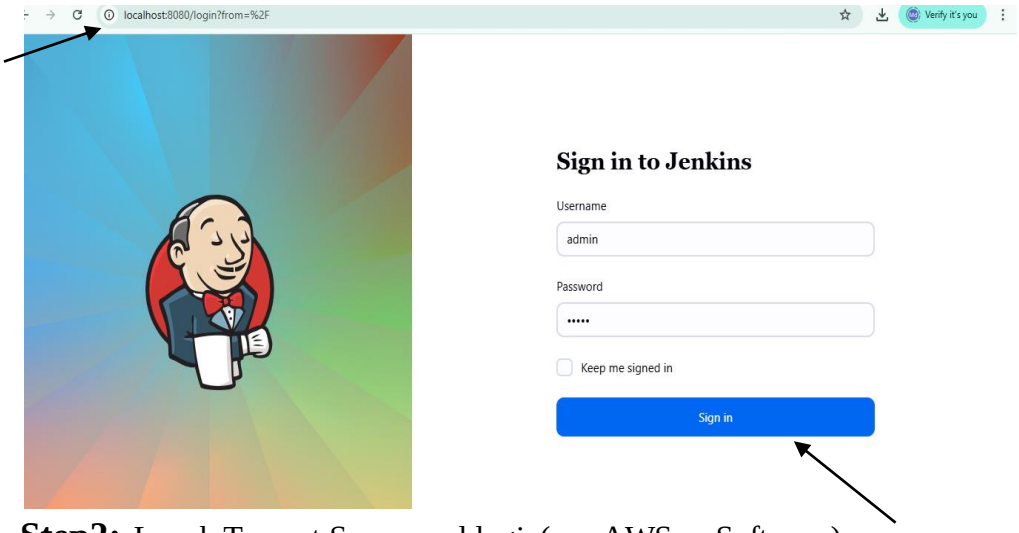
AIM:-Perform Automation using Jenkins.

Softwares:Jenkins(port:8080),JDk 17 or 21,Tomcat 9 (port:9090),Gitbash

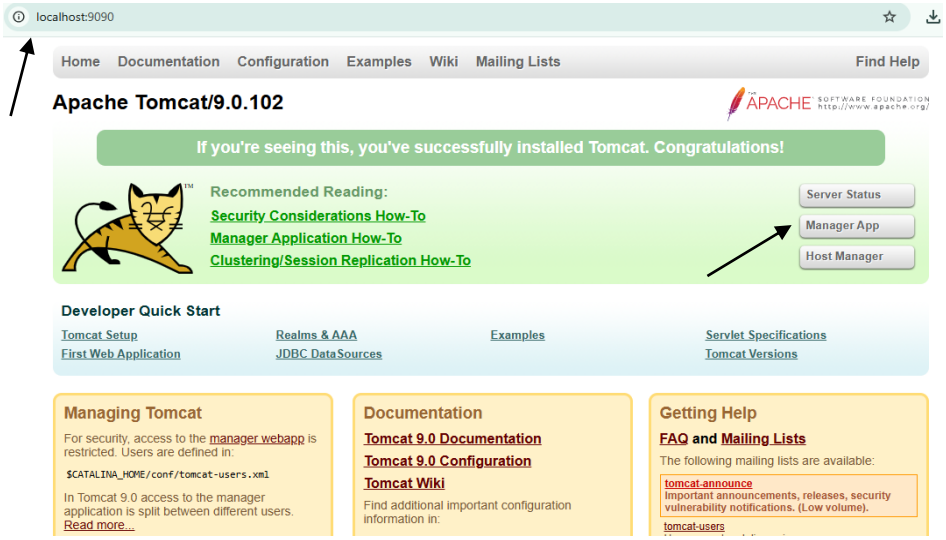
(Here we are creating one artifact war file. That war file we are deploy to test server(Tomcat server) using Jenkins Server)

(**Note:-**Images are purpose of understanding only don't paste it in records but output images paste it in output area)

Step1:-Lunch Jenkins and Login(Use AWS or Software)



Step2:-Lunch Tomcat Server and login(use AWS or Software)



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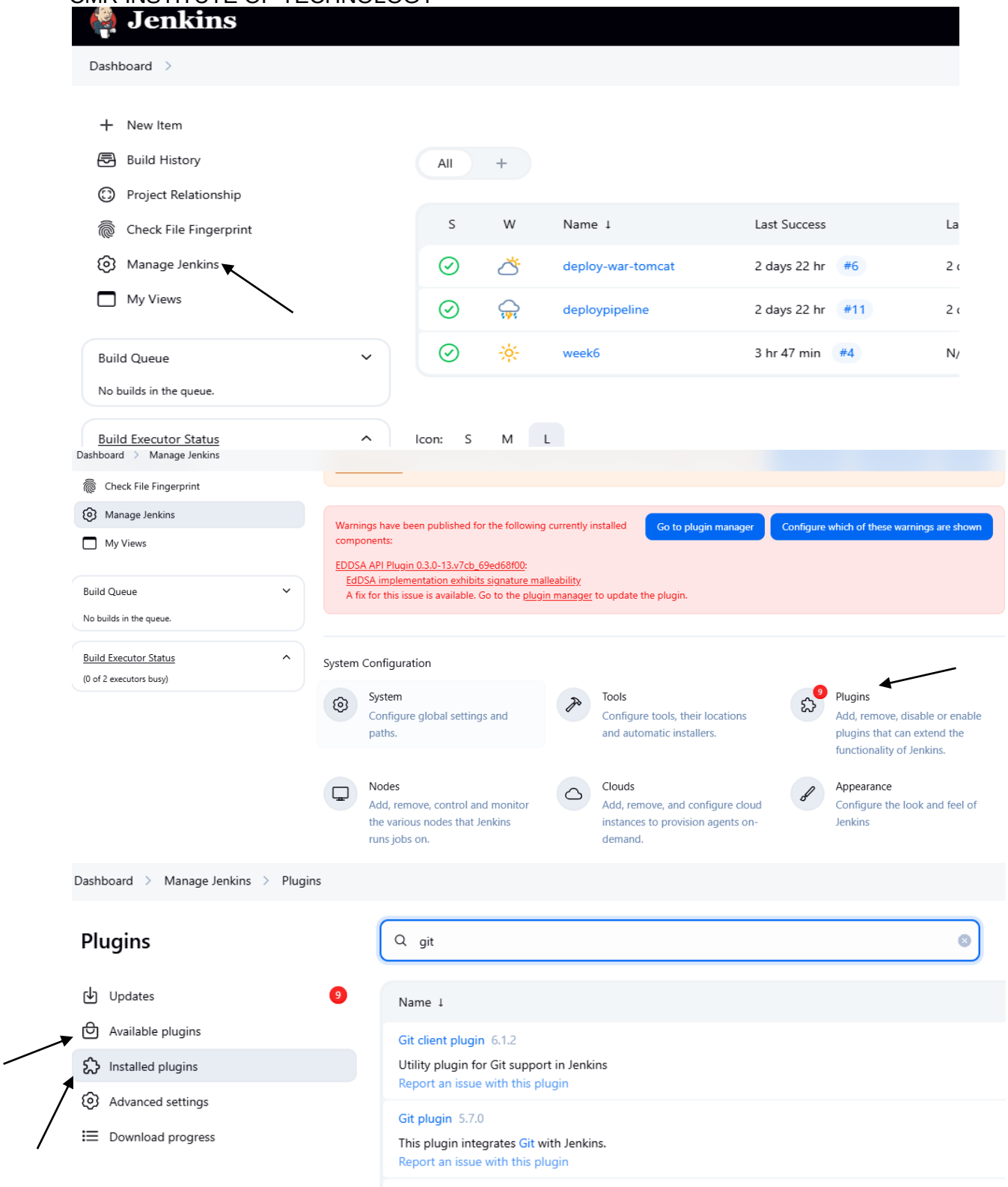

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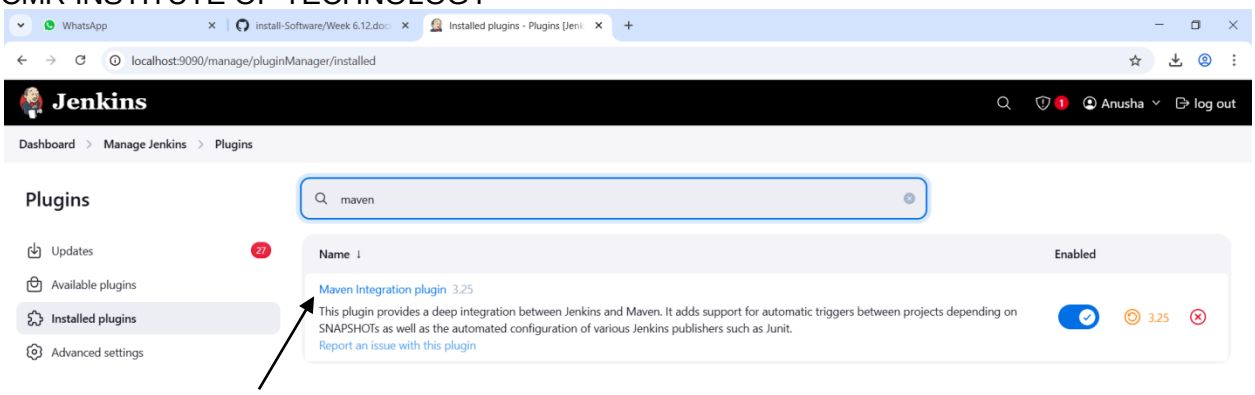

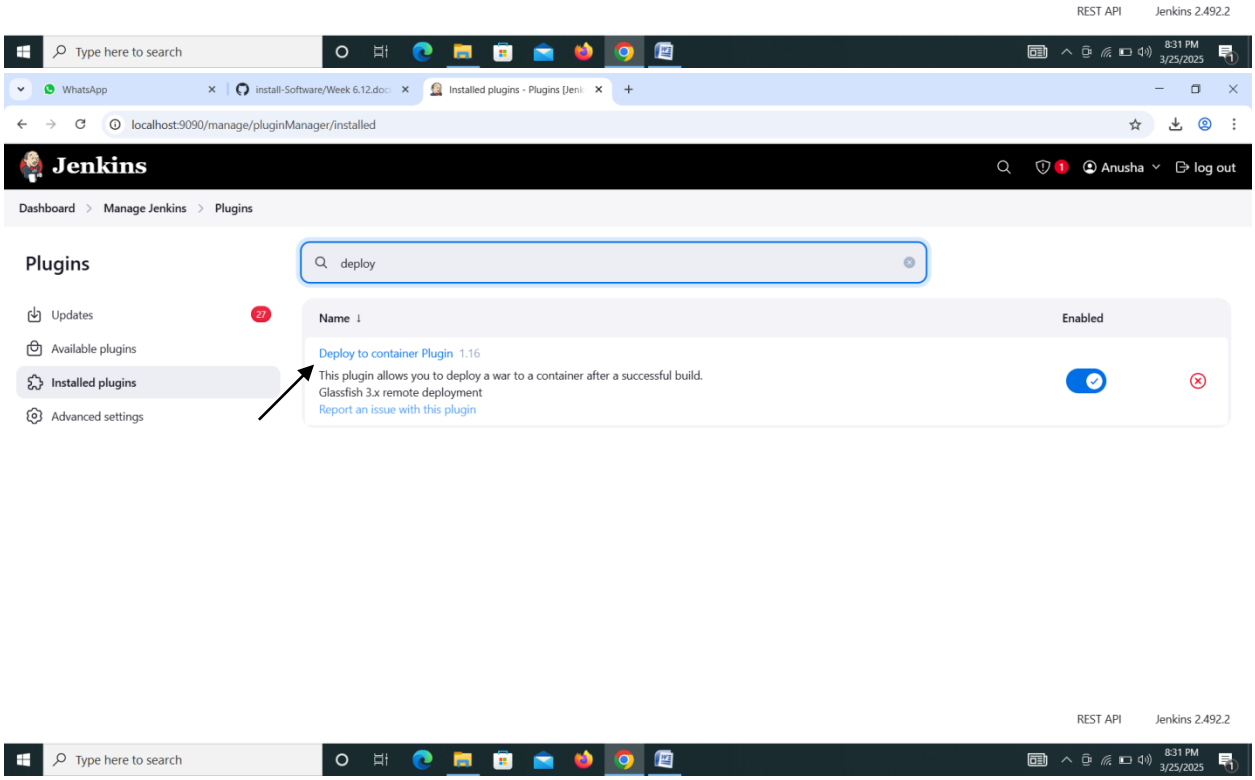

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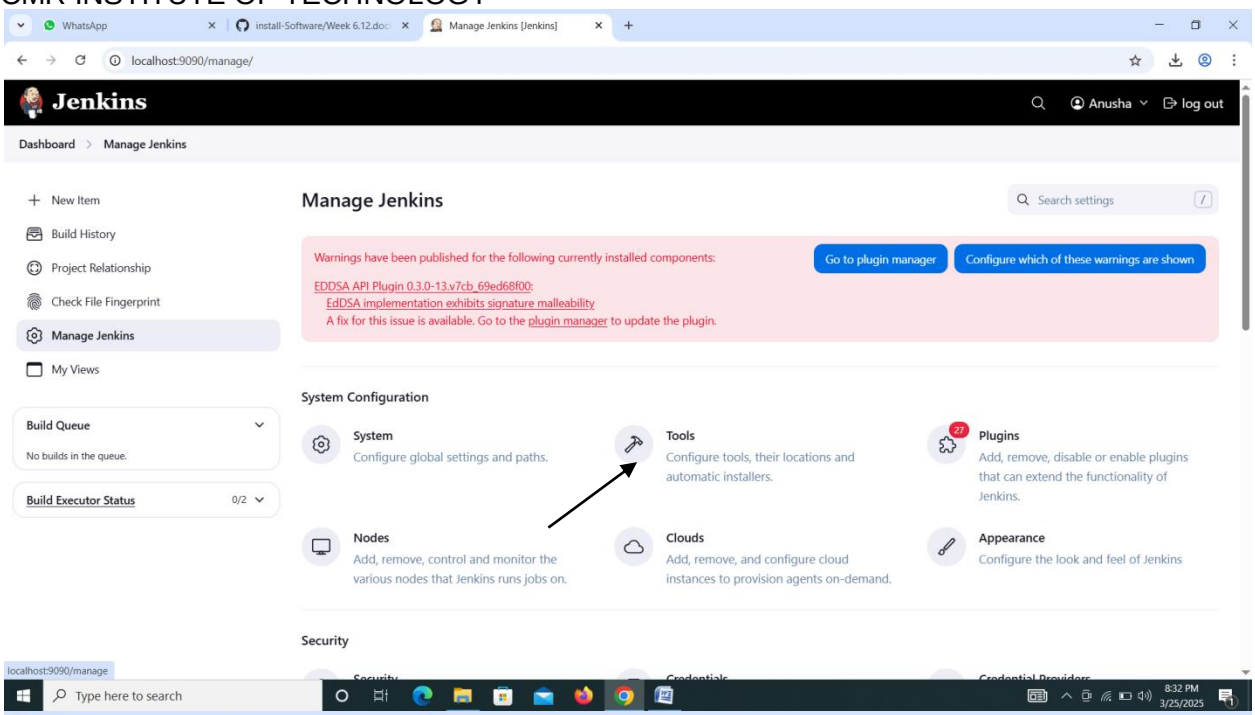
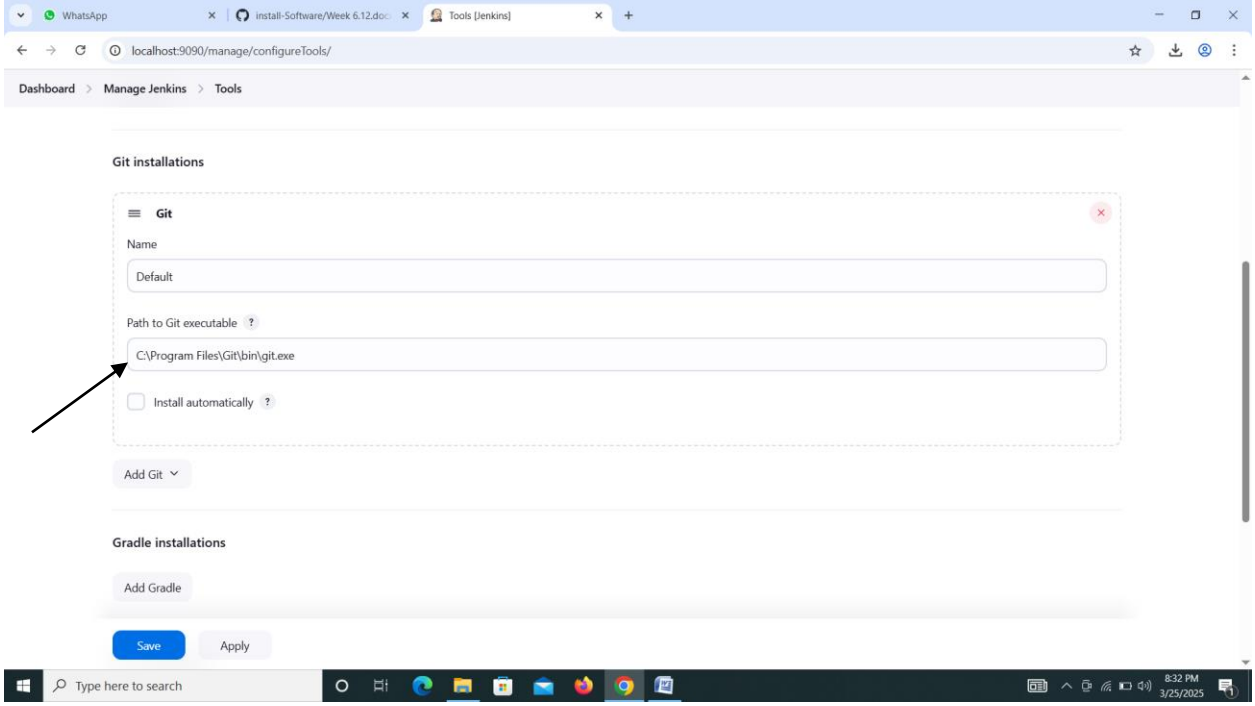
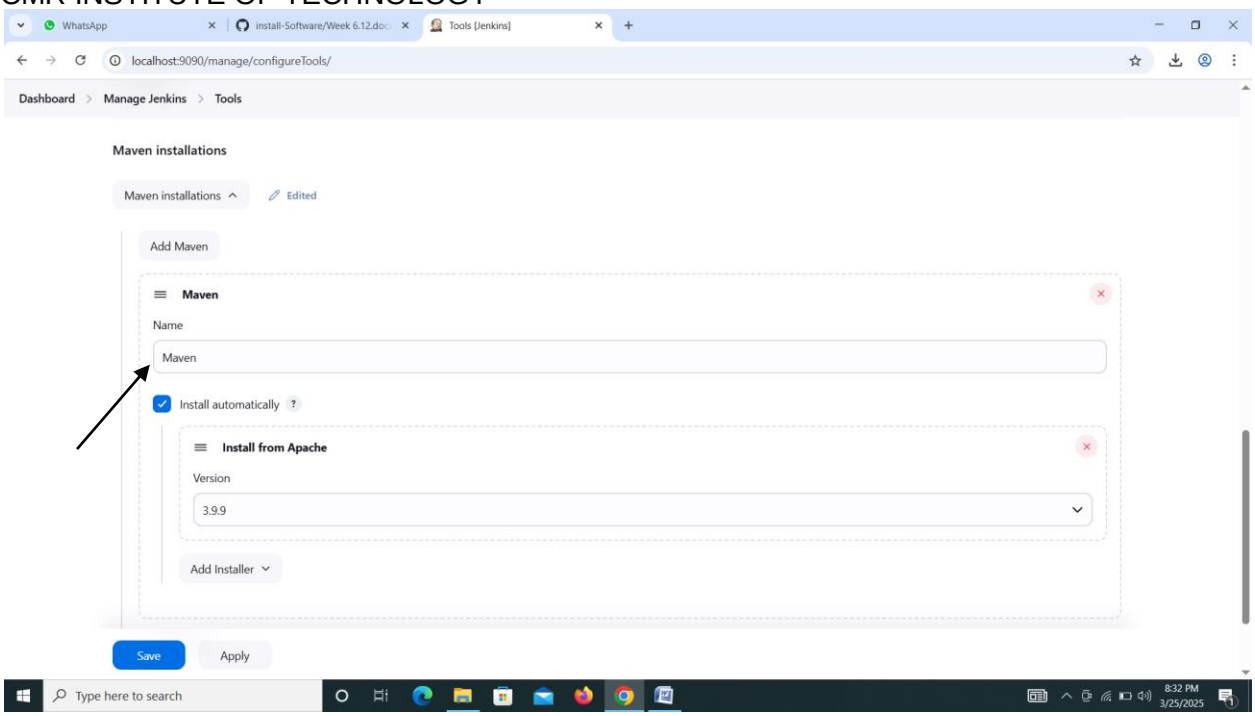
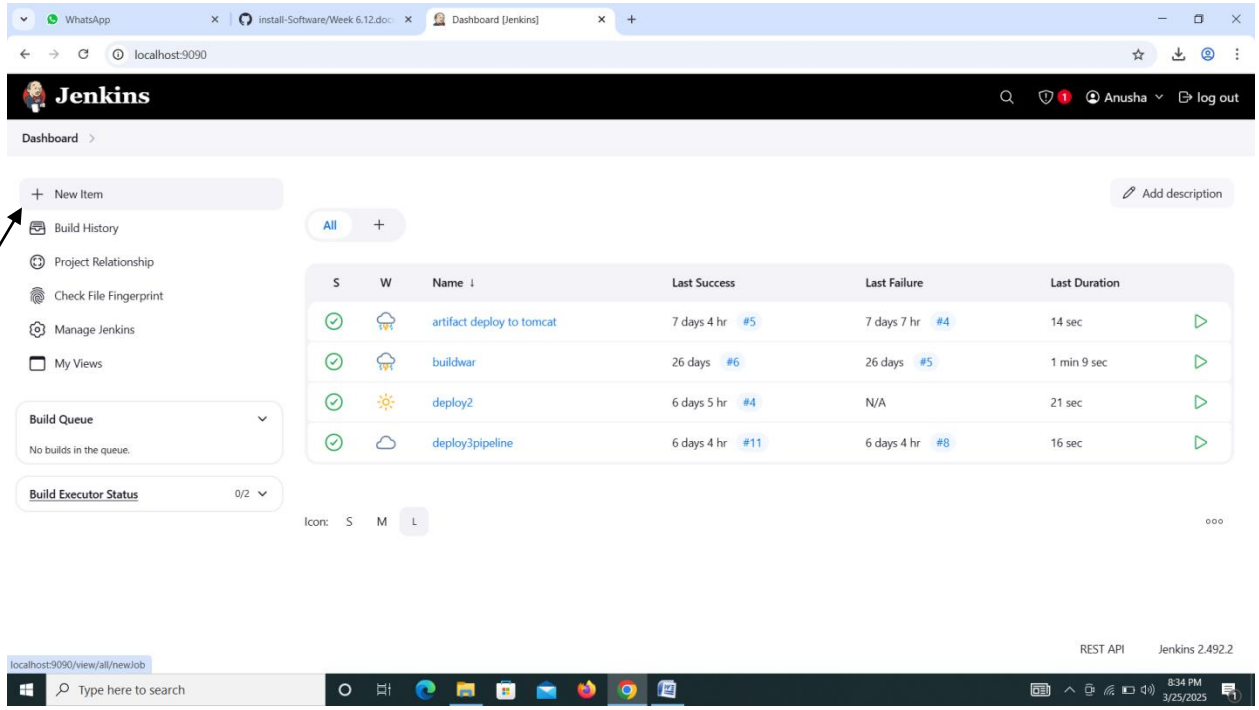

Manage Jenkins [Jenkins]
localhost:9090/manage/
Jenkins
Anusha
log out
Dashboard > Manage Jenkins
New Item
Build History
Project Relationship
Check File Fingerprint
Manage Jenkins
My Views
Build Queue
No builds in the queue.
Build Executor Status 0/2
Manage Jenkins
Search settings
Warnings have been published for the following currently installed components:
EDDSA API Plugin 0.3.0-13.v7cb_69ed68f00:
EdDSA implementation exhibits signature malleability
A fix for this issue is available. Go to the plugin manager to update the plugin.
Go to plugin manager
Configure which of these warnings are shown
System Configuration
System
Configure global settings and paths.
Tools
Configure tools, their locations and automatic installers.
Plugins
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.
Nodes
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
Clouds
Add, remove, and configure cloud instances to provision agents on-demand.
Appearance
Configure the look and feel of Jenkins
Security

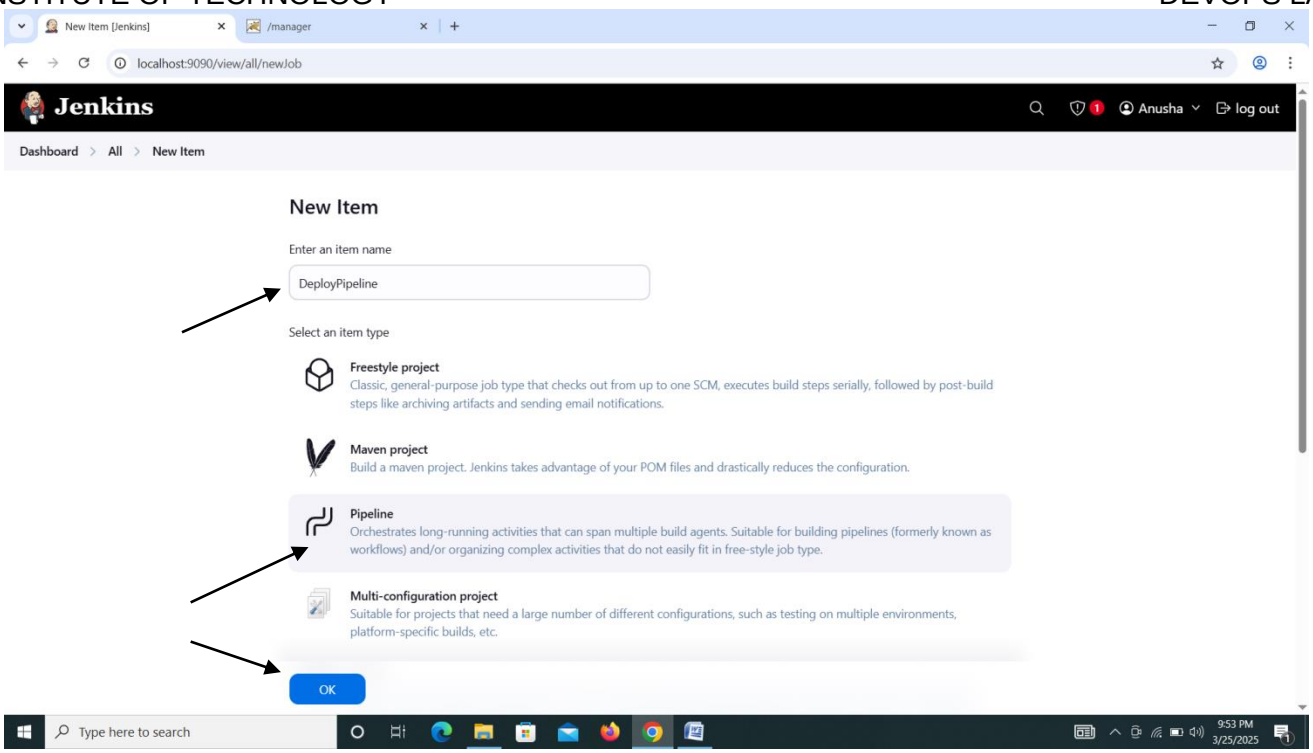

Tools [Jenkins]
localhost:9090/manage/configureTools/
Dashboard > Manage Jenkins > Tools
Git installations
Git
Name
Default
Path to Git executable
C:\Program Files\Git\bin\git.exe
Install automatically
Add Git
Gradle installations
Add Gradle
Save
Apply



Step 5:-Creating New Job for deploy artifact in tomcat test server

6. Create New Item→ Enter name DeployPipeline →select pipeline project→click ok

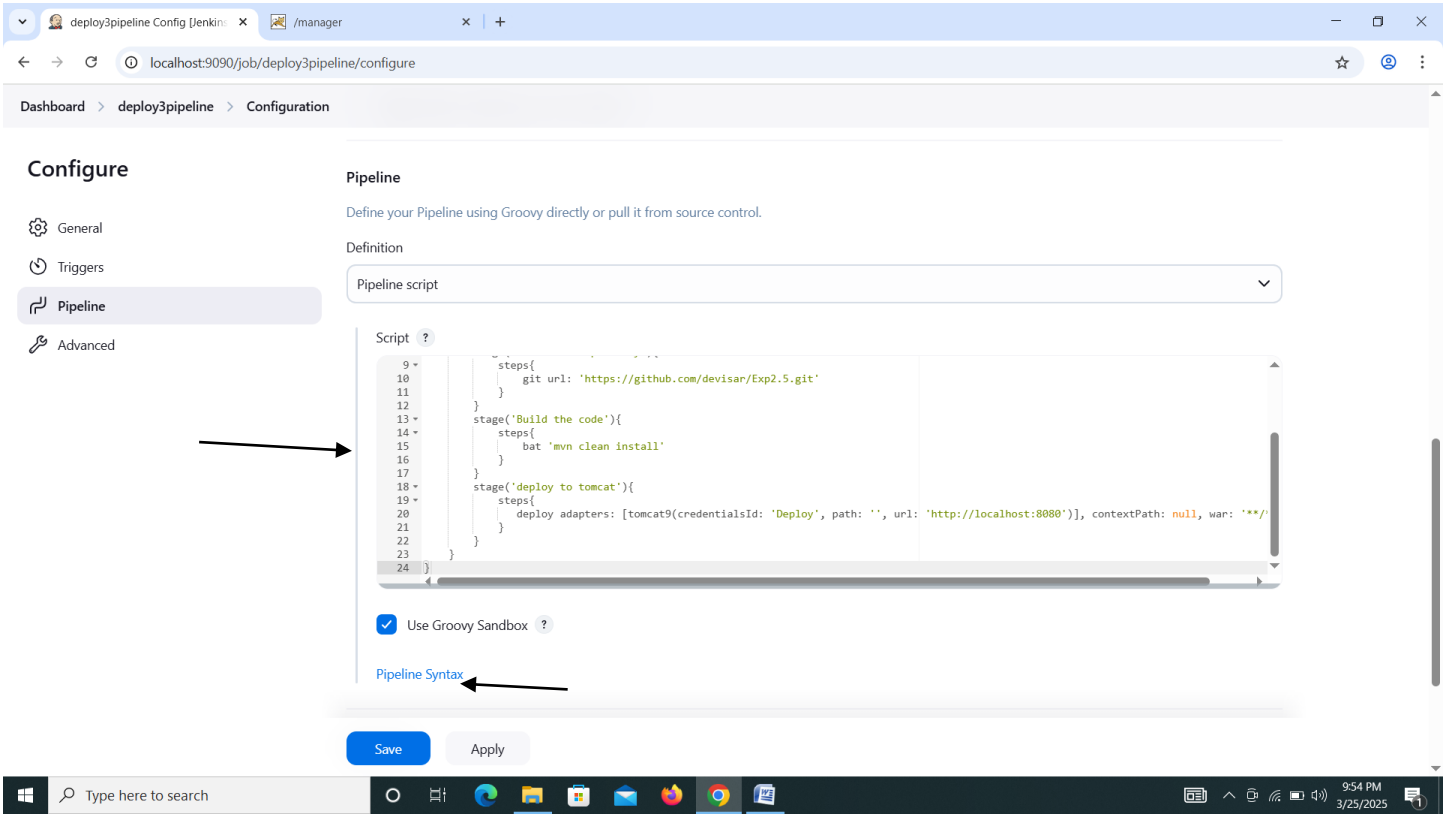




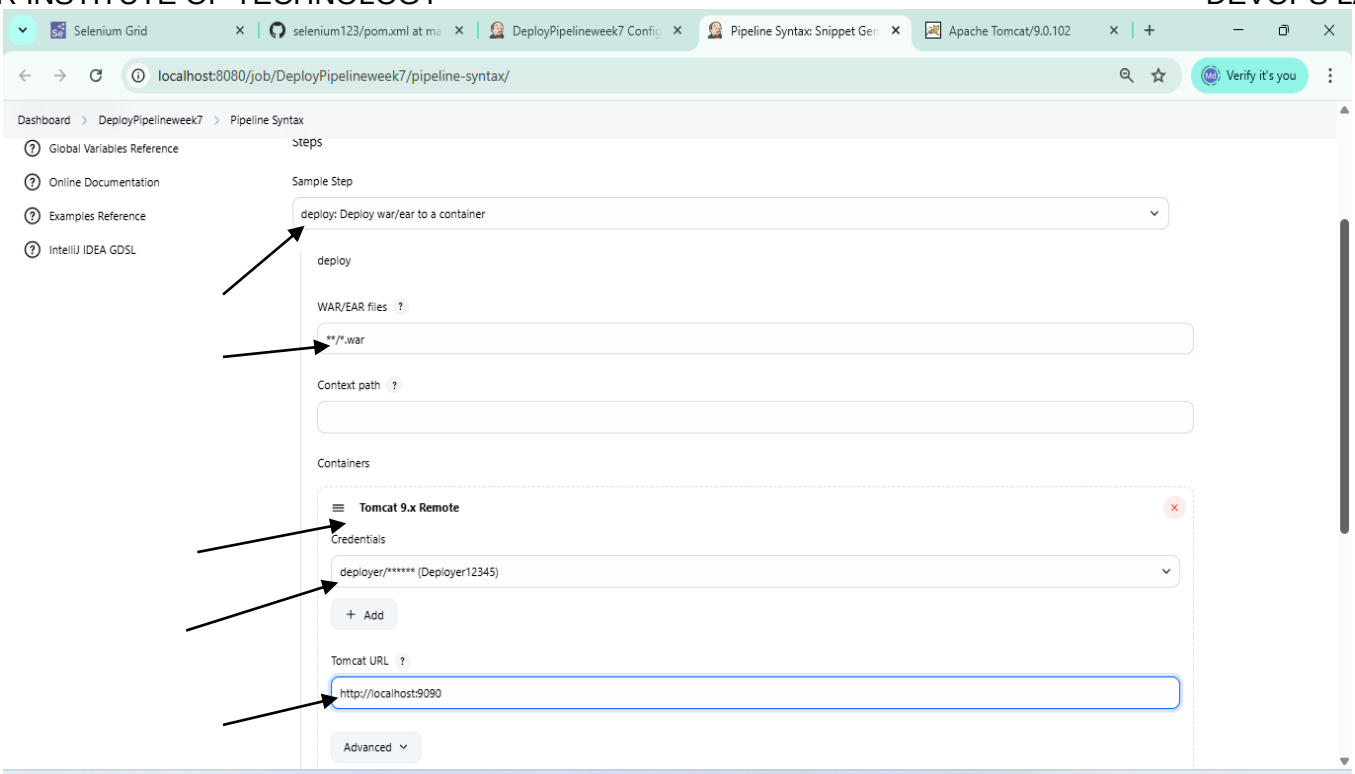
Step 2:-Copy paste below commands pipeline script

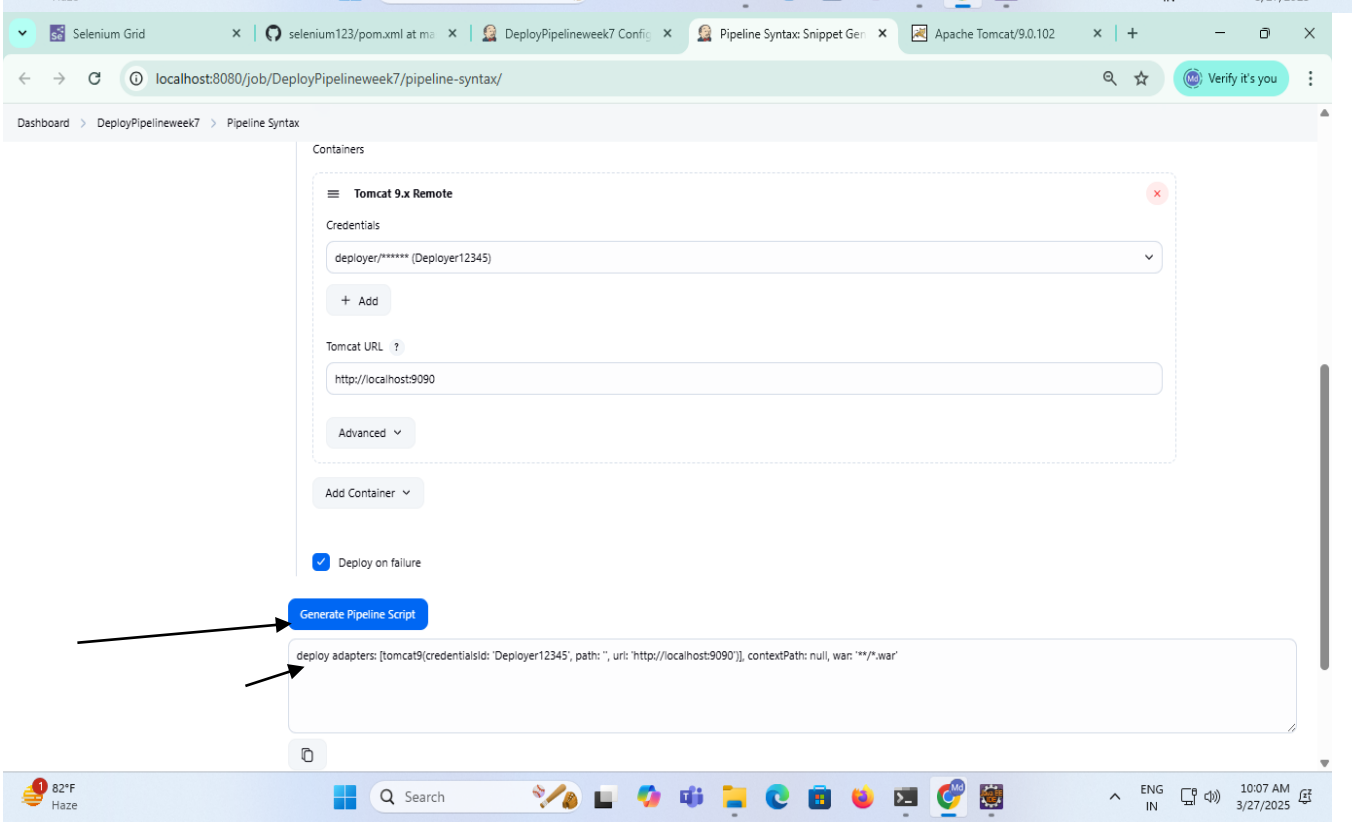
```
pipeline{
   agent any
   tools{
      maven 'Maven'

   }
   stages{
      stage('clone the repository'){
         steps{
            git url: 'https://github.com/devisar/Exp2.5.git'
         }
      }
      stage('Build the code'){
         steps{
            bat 'mvn clean install'
         }
      }
      stage('deploy to tomcat'){
         steps{
           deploy adapters: [tomcat9(credentialsId: 'Deploy', path: '', url: 'http://localhost:8080')],
contextPath: null, war: '**/*.war'
         }
      }
```

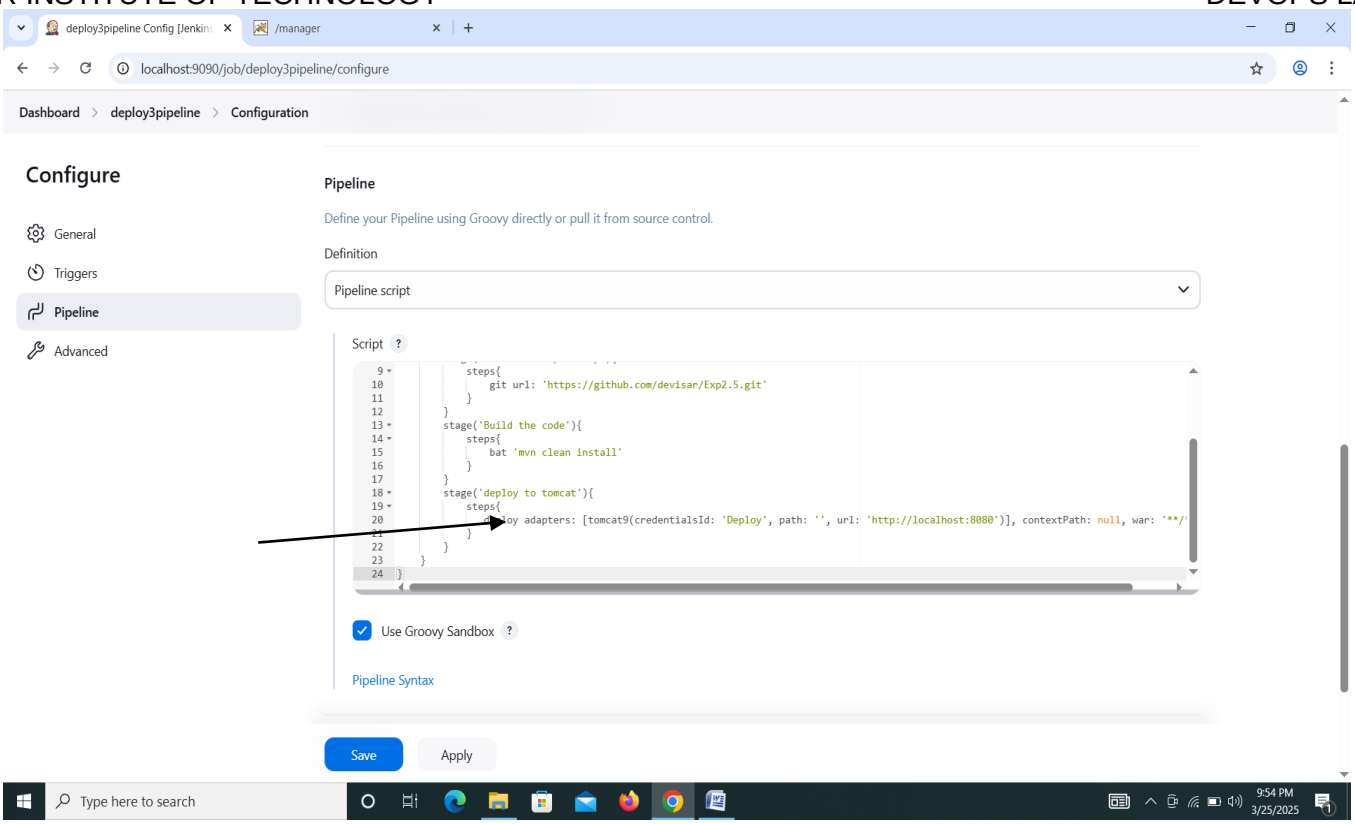
```
    }
}
```



Step 6:-deploy to tomcat you want to create pipeline script go to →pipeline syntax→Select in sample step deploy: Deploy war/ear to a container →enter war/ear files:**/*.war→Select Containers tomcat 9→select credentials tomcat user name and password. →Enter Tomcat Url like http://localhost:9090. →Click Generate Pipeline Script copy and paste it original pipeline deployment stage.

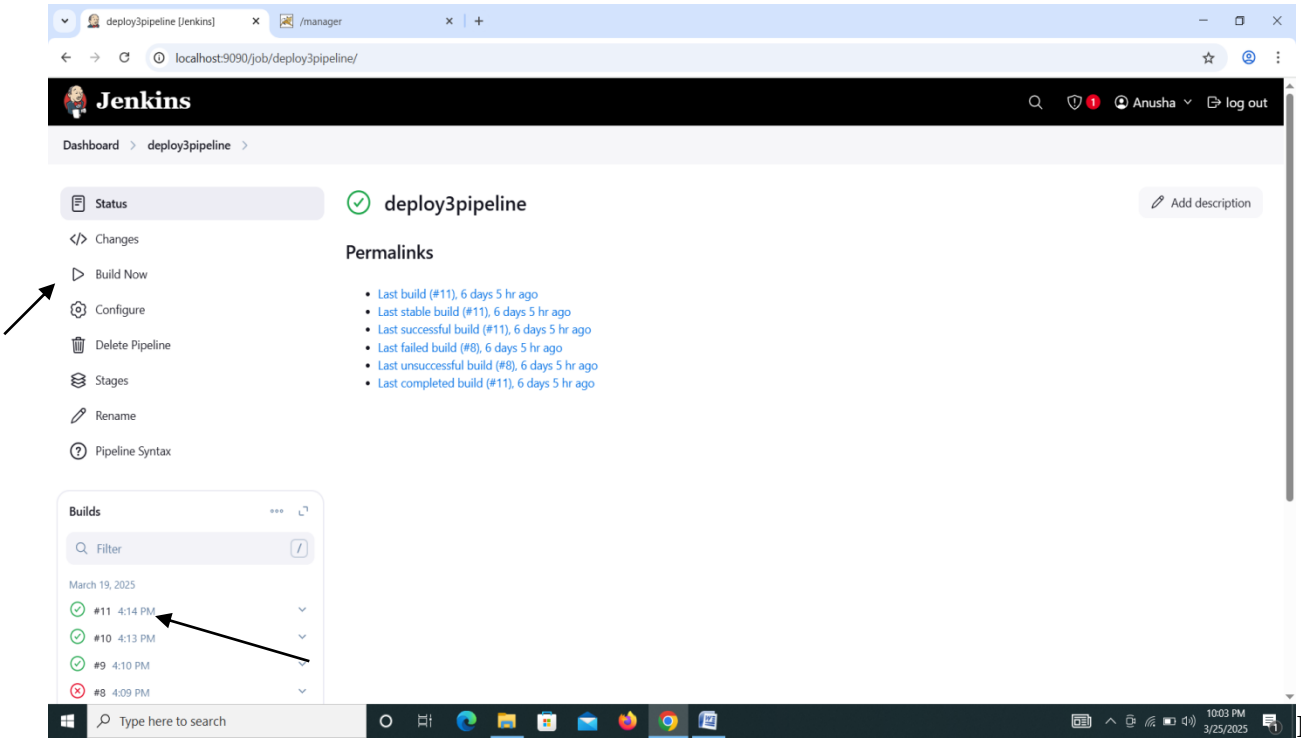

Selenium Grid
selenium123/pom.xml at ma
DeployPipelineweek7 Config
Pipeline Syntax: Snippet Gen
Apache Tomcat/9.0.102
localhost:8080/job/DeployPipelineweek7/pipeline-syntax/
Verify it's you
Dashboard > DeployPipelineweek7 > Pipeline Syntax
Global Variables Reference
Online Documentation
Examples Reference
IntelliJ IDEA GDSL
Steps
Sample Step
deploy: Deploy war/ear to a container
deploy
WAR/EAR files
**/*.war
Context path
Containers
Tomcat 9.x Remote
Credentials
deployer/****** (Deployer12345)
+ Add
Tomcat URL
http://localhost:9090
Advanced
82°F Haze
Search
ENG IN
10:07 AM 3/27/2025


Selenium Grid
selenium123/pom.xml at ma
DeployPipelineweek7 Config
Pipeline Syntax: Snippet Gen
Apache Tomcat/9.0.102
localhost:8080/job/DeployPipelineweek7/pipeline-syntax/
Verify it's you
Dashboard > DeployPipelineweek7 > Pipeline Syntax
Containers
Tomcat 9.x Remote
Credentials
deployer/****** (Deployer12345)
+ Add
Tomcat URL
http://localhost:9090
Advanced
Add Container
Deploy on failure
Generate Pipeline Script
deploy adapters: [tomcat9(credentialsId: 'Deployer12345', path: '', url: 'http://localhost:9090')], contextPath: null, war: '**/*.war'
82°F Haze
Search
ENG IN
10:07 AM 3/27/2025

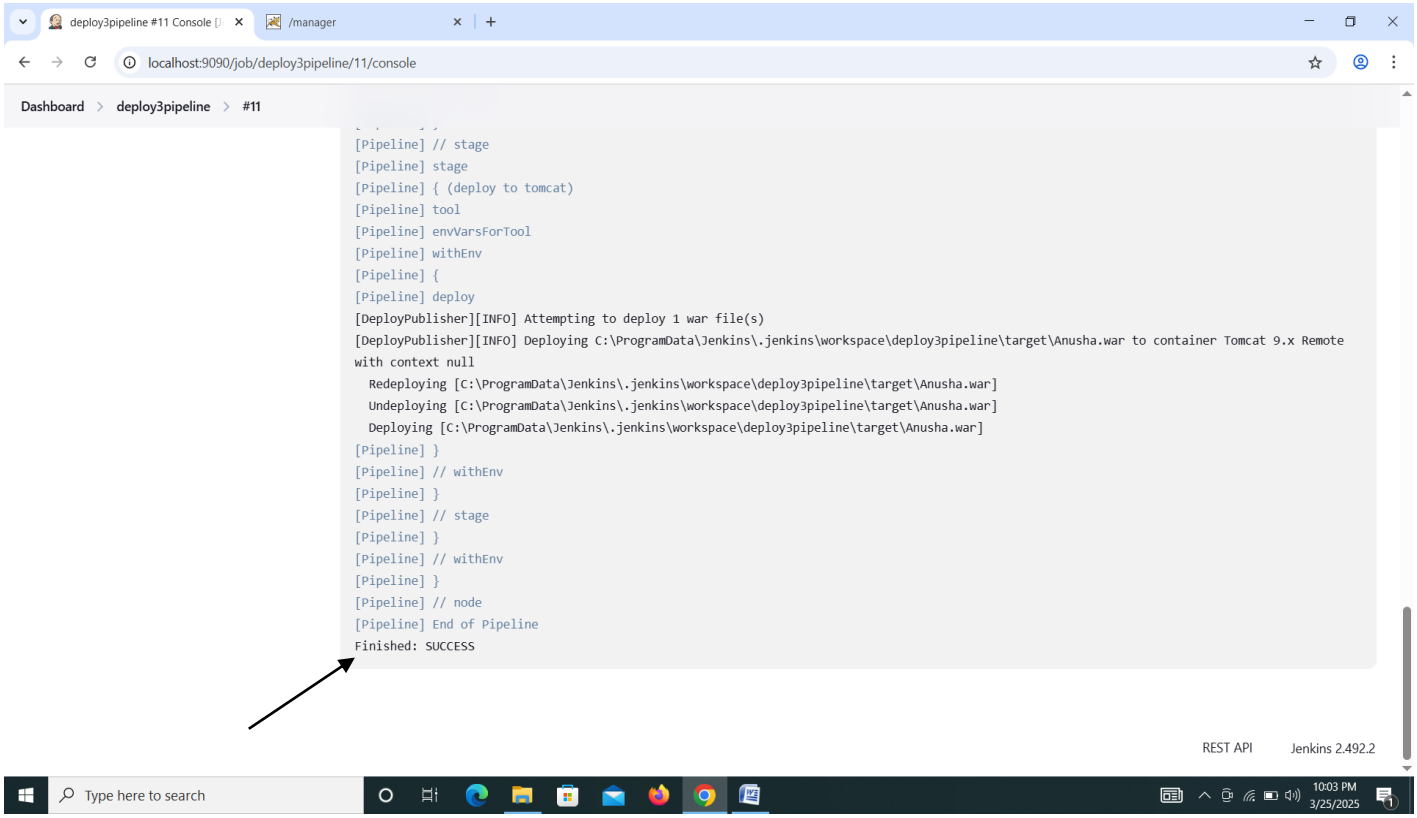


Step 7:- Build war File

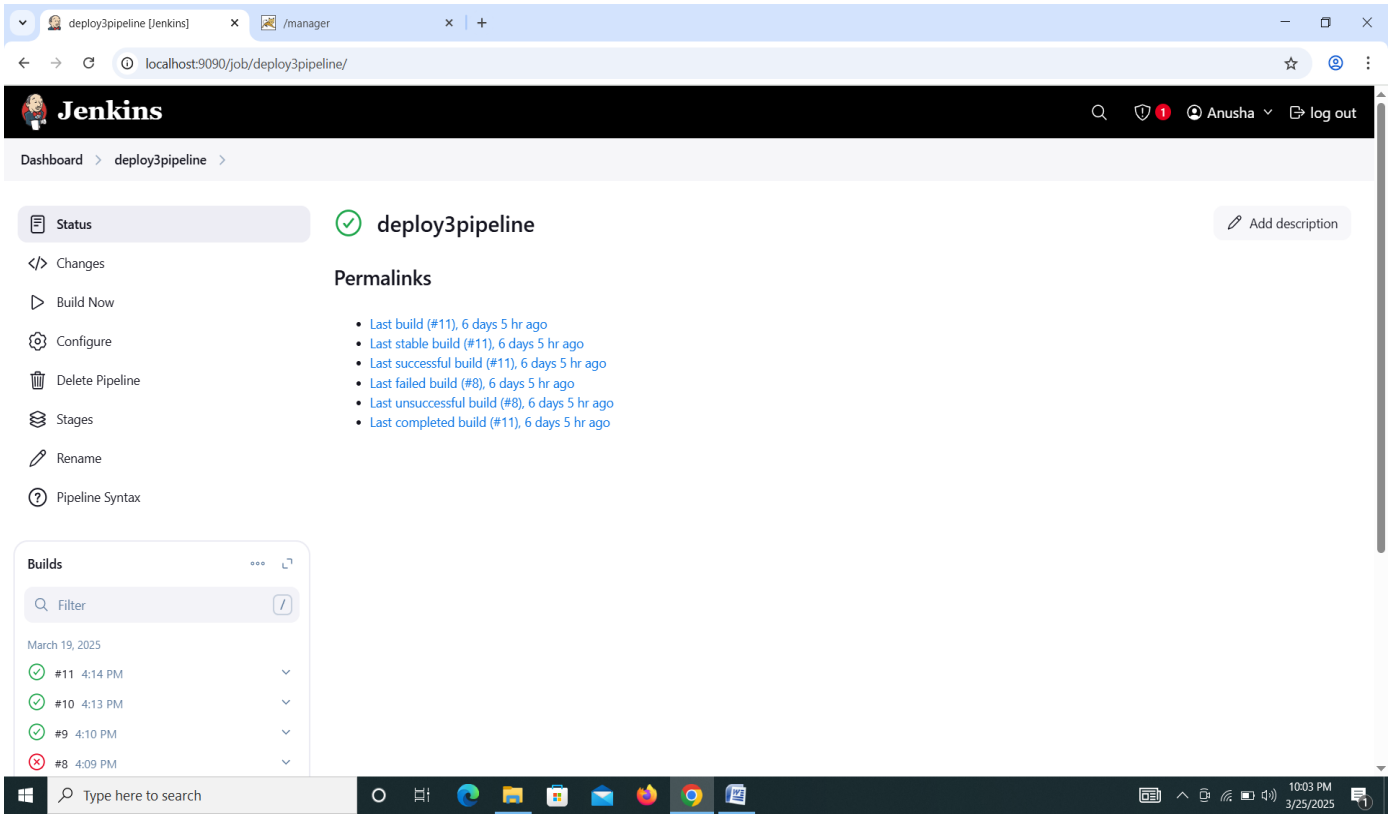
2. Click Build now →Go to

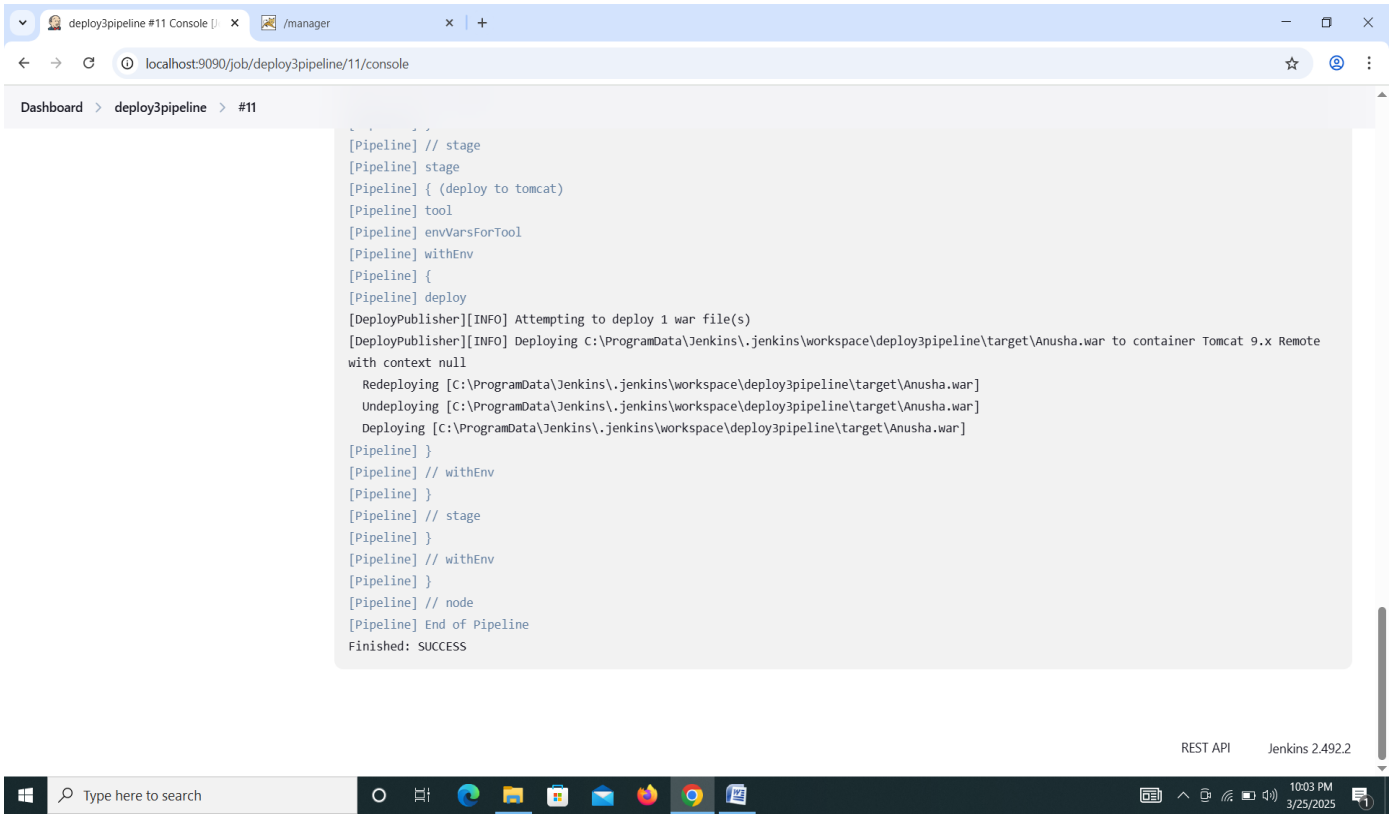


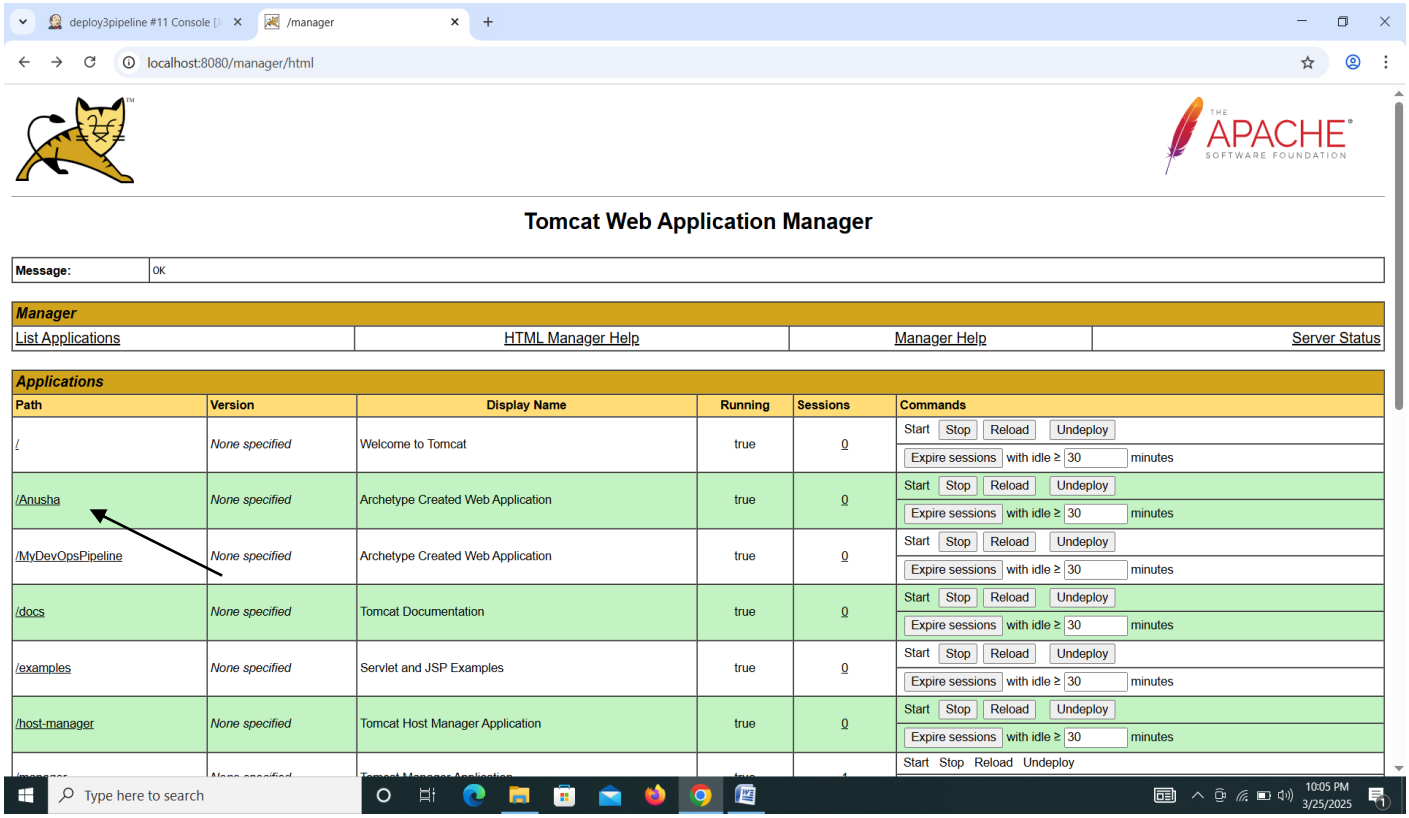
Build select console output now you got Successfully Finish.

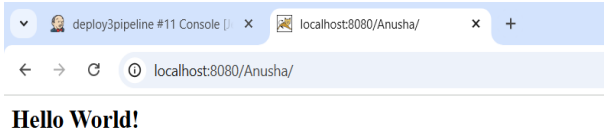

deploy3pipeline #11 Console
/manager
localhost:9090/job/deploy3pipeline/11/console
Dashboard > deploy3pipeline > #11
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (deploy to tomcat)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] deploy
[DeployPublisher][INFO] Attempting to deploy 1 war file(s)
[DeployPublisher][INFO] Deploying C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war to container Tomcat 9.x Remote with context null
Redeploying [C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war]
Undeploying [C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war]
Deploying [C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war]
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
REST API
Jenkins 2.492.2
Type here to search
10:03 PM
3/25/2025

Output:-


deploy3pipeline [Jenkins]
/manager
localhost:9090/job/deploy3pipeline/
Jenkins
Anusha
log out
Dashboard > deploy3pipeline >
Status
Changes
Build Now
Configure
Delete Pipeline
Stages
Rename
Pipeline Syntax
deploy3pipeline
Add description
Permalinks
Last build (#11), 6 days 5 hr ago
Last stable build (#11), 6 days 5 hr ago
Last successful build (#11), 6 days 5 hr ago
Last failed build (#8), 6 days 5 hr ago
Last unsuccessful build (#8), 6 days 5 hr ago
Last completed build (#11), 6 days 5 hr ago
Builds
Filter
March 19, 2025
#11 4:14 PM
#10 4:13 PM
#9 4:10 PM
#8 4:09 PM
Type here to search
10:03 PM
3/25/2025


deploy3pipeline #11 Console
/manager
localhost:9090/job/deploy3pipeline/11/console
Dashboard > deploy3pipeline > #11
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (deploy to tomcat)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] deploy
[DeployPublisher][INFO] Attempting to deploy 1 war file(s)
[DeployPublisher][INFO] Deploying C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war to container Tomcat 9.x Remote
with context null
Redeploying [C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war]
Undeploying [C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war]
Deploying [C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war]
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
REST API
Jenkins 2.492.2
Type here to search
10:03 PM
3/25/2025


deploy3pipeline #11 Console
/manager
localhost:8080/manager/html
Tomcat Web Application Manager
Message: OK
Manager
List Applications
HTML Manager Help
Manager Help
Server Status
Applications
Path
Version
Display Name
Running
Sessions
Commands
/
None specified
Welcome to Tomcat
true
0
/Anusha
None specified
Archetype Created Web Application
true
0
/MyDevOpsPipeline
None specified
Archetype Created Web Application
true
0
/docs
None specified
Tomcat Documentation
true
0
/examples
None specified
Servlet and JSP Examples
true
0
/host-manager
None specified
Tomcat Host Manager Application
true
0
Start Stop Reload Undeploy
Expire sessions with idle ≥ 30 minutes
Type here to search
10:05 PM
3/25/2025


deploy3pipeline #11 Console
localhost:8080/Anusha/
localhost:8080/Anusha/
Hello World!